

Reading Notes: Analytic Combinatorics

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Chapter 1

COMBINATORIAL STRUCTURES AND ORDINARY GENERATING FUNCTIONS

1.1 Symbolic enumeration methods

▷ **I.2.** *Unimodal permutations.*

A unimodal permutation rises to a single summit, namely n , and then comes back down. So the whole shape is: an increasing run, the peak n , and then a decreasing run. This class admits the symbolic specification

$$\mathcal{P}^U = \text{SET}(\mathcal{Z}) \star \mathcal{Z}^\blacksquare \star \text{SET}(\mathcal{Z})$$

Accordingly, its exponential generating function is

$$P^U(z) = \int_0^z \left(e^t \cdot \frac{d}{dt} t \cdot e^t \right) dt = \frac{1}{2}(e^{2z} - 1)$$

and therefore

$$P_n^U = n! [z^n] \frac{1}{2} e^{2z} = n! \frac{1}{2} \frac{2^n}{n!} = 2^{n-1}.$$

There is also a one-line counting proof. If n sits in position k , then we only need to choose which $k - 1$ of the remaining $n - 1$ entries go to its left; once that choice is made, the left-hand side must be in increasing order and the right-hand side in decreasing order. Hence

$$P_n^U = \sum_{k=1}^n \binom{n-1}{k-1} = \sum_{j=0}^{n-1} \binom{n-1}{j} = 2^{n-1}.$$

1.2 Admissible constructions and specifications

1.3 Integer compositions and partitions

▷ **I.16.** *Partitions with summands bounded in number and size.*

Consider partitions of size n with at most k summands, each bounded by l . The corresponding counting formula is

$$[z^n] \frac{(1-z)(1-z^2) \cdots (1-z^{k+l})}{((1-z)(1-z^2) \cdots (1-z^k)) \cdot ((1-z)(1-z^2) \cdots (1-z^l))} \quad (1.1)$$

Let $f(k, l)$ denote the rational function inside (1.1), so that the coefficient of z^n counts the admissible partitions of n . The natural split is by the largest part: it is either strictly smaller than l , or exactly l . Thus

$$f(k, l) = f(k, l-1) + z^l f(k-1, l) \quad (1.2)$$

To package the dependence on l , write

$$F_k(x) = \sum_{l=0}^{\infty} f(k, l) x^l$$

For the base case $k = 0$, only the empty partition survives, so $f(0, l) = 1$ for all $l \geq 0$. Hence

$$F_0(x) = \sum_{l=0}^{\infty} 1 \cdot x^l = \frac{1}{1-x}$$

Now multiply (1.2) by x^l and sum over $l \geq 1$:

$$\sum_{l=1}^{\infty} f(k, l) x^l - \sum_{l=1}^{\infty} f(k, l-1) x^l = \sum_{l=1}^{\infty} f(k-1, l) (zx)^l$$

- The first term simplifies to $F_k(x) - 1$, since $f(k, 0) = 1$.
- The second term is simply $x F_k(x)$.
- The right-hand side yields $F_{k-1}(zx) - 1$.

Substituting these pieces back in gives

$$(1-x)F_k(x) - 1 = F_{k-1}(zx) - 1 \implies F_k(x) = \frac{1}{1-x} F_{k-1}(zx)$$

Iterating the relation gives

$$\begin{aligned} F_k(x) &= \frac{1}{1-x} F_{k-1}(zx) \\ &= \frac{1}{1-x} \frac{1}{1-zx} F_{k-2}(z^2x) \\ &= \frac{1}{(1-x)(1-zx) \cdots (1-z^{k-1}x)} F_0(z^k x) \end{aligned}$$

Since $F_0(x) = \frac{1}{1-x}$, we have $F_0(z^k x) = \frac{1}{1-z^k x}$. Thus

$$F_k(x) = \sum_{l=0}^{\infty} f(k, l) x^l = \frac{1}{(1-x)(1-zx) \cdots (1-z^k x)}$$

To identify this generating function more explicitly, we now decompose it into partial fractions:

$$\frac{1}{\prod_{i=0}^k (1 - z^i x)} = \sum_{j=0}^k \frac{A_j}{1 - z^j x}$$

To isolate A_j , multiply through by $(1 - z^j x)$ and then set $x = z^{-j}$:

$$\begin{aligned} A_j &= \left[\frac{1 - z^j x}{\prod_{i=0}^k (1 - z^i x)} \right]_{x=z^{-j}} \\ &= \frac{1}{\prod_{i=0}^{j-1} (1 - z^i z^{-j}) \prod_{i=j+1}^k (1 - z^i z^{-j})} \\ &= \frac{(-1)^j z^{j(j+1)/2}}{\prod_{i=1}^j (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \end{aligned}$$

The coefficient of x^l in $F_k(x)$ is then read off term by term. For $\frac{A_j}{1 - z^j x}$, it is simply $A_j(z^j)^l$.

$$\begin{aligned} f(k, l) &= \sum_{j=0}^k A_j (z^j)^l \\ &= \sum_{j=0}^k \frac{(-1)^j z^{j(j+1)/2} \cdot z^{jl}}{\prod_{i=1}^j (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \\ &= \frac{1}{\prod_{i=1}^k (1 - z^i)} \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j+1)/2 + jl} \end{aligned}$$

where $\binom{k}{j}_z$ is the Gaussian binomial coefficient:

$$\begin{aligned} \binom{k}{j}_z &= \frac{\prod_{i=1}^k (1 - z^i)}{\prod_{i=1}^j (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \\ &= \frac{(1 - z^k)(1 - z^{k-1}) \cdots (1 - z^{k-j+1})}{(1 - z)(1 - z^2) \cdots (1 - z^j)} \end{aligned}$$

A small rearrangement of the exponent gives

$$\frac{j(j+1)}{2} + jl = \frac{j(j-1)}{2} + j(l+1)$$

so the sum becomes

$$f(k, l) = \frac{1}{\prod_{i=1}^k (1 - z^i)} \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j-1)/2} (z^{l+1})^j$$

The q -binomial theorem states that for a variable X :

$$\prod_{i=0}^{k-1} (1 - z^i X) = \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j \quad (1.3)$$

Substituting $X = z^{l+1}$ into the theorem recovers (1.1):

$$\begin{aligned} f(k, l) &= \frac{(1 - z^{l+1})(1 - z^{l+2}) \cdots (1 - z^{l+k})}{(1 - z^1)(1 - z^2) \cdots (1 - z^k)} \\ &= \frac{(1 - z)(1 - z^2) \cdots (1 - z^{k+l})}{((1 - z)(1 - z^2) \cdots (1 - z^k)) \cdot ((1 - z)(1 - z^2) \cdots (1 - z^l))} \end{aligned}$$

At this point, the original product formula has been recovered. The rest is optional background, but it is pleasant enough to record: let us also justify (1.3). Set

$$P_k(X) = \prod_{i=0}^{k-1} (1 - z^i X)$$

We claim that $P_k(X) = \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$. The useful relation comes from comparing $P_k(zX)$ with $P_k(X)$:

$$P_k(zX) = \prod_{i=0}^{k-1} (1 - z^{i+1} X) = \frac{1 - z^k X}{1 - X} P_k(X)$$

After rearranging, this becomes the functional equation

$$(1 - X)P_k(zX) = (1 - z^k X)P_k(X)$$

Let $P_k(X) = \sum_{j=0}^k c_j X^j$. We substitute this power series into the functional equation:

$$(1 - X) \sum_{j=0}^k c_j (zX)^j = (1 - z^k X) \sum_{j=0}^k c_j X^j$$

Comparing coefficients of X^j on both sides yields:

$$c_j z^j - c_{j-1} z^{j-1} = c_j - c_{j-1} z^k$$

so

$$c_j = c_{j-1} \frac{z^{j-1} - z^k}{z^j - 1} = -c_{j-1} z^{j-1} \frac{1 - z^{k-j+1}}{1 - z^j}$$

Starting from $c_0 = P_k(0) = 1$, we iterate:

$$c_1 = -c_0 z^0 \frac{1 - z^k}{1 - z^1}, \quad c_2 = -c_1 z^1 \frac{1 - z^{k-1}}{1 - z^2} = (-1)^2 z^{0+1} \frac{(1 - z^k)(1 - z^{k-1})}{(1 - z^1)(1 - z^2)}$$

The exponent of z is $0 + 1 + 2 + \cdots + (j-1) = \frac{j(j-1)}{2}$, and the remaining fraction is exactly the Gaussian binomial coefficient. Therefore,

$$c_j = (-1)^j z^{j(j-1)/2} \binom{k}{j}_z$$

Plugging the coefficients back into the expansion, we obtain

$$\prod_{i=0}^{k-1} (1 - z^i X) = \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$$

as claimed.

▷ **I.17.** *Systems of linear Diophantine inequalities.*

This note contains five counting problems. They share a common theme, but each has its own constraint set and can be solved independently.

Problem 1

$$\{x_1 \geq 0, x_2 \geq 2x_1, x_3 \geq 2x_2, x_4 \geq 2x_3\}$$

At first glance this system looks a little forbidding, but slack variables settle it quickly. Introduce non-negative variables $y_j \in \mathbb{Z}_{\geq 0}$ to measure the excess above each lower bound.

$$\begin{aligned} x_1 &= y_1 \\ x_2 &= 2x_1 + y_2 = 2y_1 + y_2 \\ x_3 &= 2x_2 + y_3 = 4y_1 + 2y_2 + y_3 \\ x_4 &= 2x_3 + y_4 = 8y_1 + 4y_2 + 2y_3 + y_4 \end{aligned}$$

The total weight is

$$S = x_1 + x_2 + x_3 + x_4 = 15y_1 + 7y_2 + 3y_3 + y_4.$$

The generating function is now immediate: the variables y_j are independent and each ranges freely over the non-negative integers, so the OGF $F(z)$ is the product of four one-variable OGFs:

- For $15y_1$: $\sum_{y_1=0}^{\infty} (z^{15})^{y_1} = \frac{1}{1-z^{15}}$
- For $7y_2$: $\sum_{y_2=0}^{\infty} (z^7)^{y_2} = \frac{1}{1-z^7}$
- For $3y_3$: $\sum_{y_3=0}^{\infty} (z^3)^{y_3} = \frac{1}{1-z^3}$
- For y_4 : $\sum_{y_4=0}^{\infty} (z^1)^{y_4} = \frac{1}{1-z^1}$

Multiplying them together gives the answer:

$$F(z) = \frac{1}{(1-z)(1-z^3)(1-z^7)(1-z^{15})}$$

Problem 2

$$\{x_1 + x_2 \leq x_3\}$$

Here slack variables again turn the inequality into an equality with independent parts. Rewrite the constraint $x_3 \geq x_1 + x_2$ by introducing a non-negative integer $y \geq 0$:

$$x_3 = x_1 + x_2 + y$$

The total size is $S = x_1 + x_2 + x_3 + x_4 = 2x_1 + 2x_2 + y + x_4$. Since the variables x_1, x_2, y, x_4 are independent and each ranges freely over the non-negative integers, the OGF is the product of the corresponding one-variable generating functions:

- For $2x_1 : \frac{1}{1-z^2}$
- For $2x_2 : \frac{1}{1-z^2}$
- For $y : \frac{1}{1-z}$
- For $x_4 : \frac{1}{1-z}$

Multiplying these factors gives

$$F(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

Problem 3

$$\{x_1 + x_2 \geq x_3\}$$

Since the inequality does not suggest a single slack variable, the cleanest route is to work with the complement. Let $U(z) = (1-z)^{-4}$ be the OGF for four unrestricted variables. The universe of compositions splits into those with $x_1 + x_2 < x_3$ and those with $x_1 + x_2 \geq x_3$. So we first count the complement class $x_1 + x_2 < x_3$ by setting $x_3 = x_1 + x_2 + 1 + y$, where $y \geq 0$. The total sum is

$$S = x_1 + x_2 + (x_1 + x_2 + 1 + y) + x_4 = 2x_1 + 2x_2 + y + x_4 + 1$$

The OGF for this complement, denoted by $C(z)$, is

$$C(z) = \frac{z}{(1-z^2)^2(1-z)^2}$$

The OGF for our constraint is then

$$F(z) = U(z) - C(z) = \frac{1}{(1-z)^4} - \frac{z}{(1-z^2)^2(1-z)^2} = \frac{1+z+z^2}{(1-z)^4(1+z)^2}$$

Problem 4

$$\{x_1 + x_2 \leq x_3 + x_4\}$$

To find the OGF for $\{x_1 + x_2 \leq x_3 + x_4\}$, combine symmetry with a simple decomposition. Let $A = x_1 + x_2$ and $B = x_3 + x_4$. We seek the OGF for compositions satisfying $A \leq B$. In the universe of all compositions (x_1, x_2, x_3, x_4) , there are three mutually exclusive cases: 1. $A < B$, 2. $A > B$, 3. $A = B$. By symmetry, cases 1 and 2 have the same OGF, since A and B are built from the same number of variables with the same weights. Denote this common OGF by $G_{<}(z)$, and let $G_{=}(z)$ be the OGF for case 3. The total unrestricted OGF is $U(z) = \frac{1}{(1-z)^4}$. Therefore,

$$U(z) = 2G_{<}(z) + G_{=}(z)$$

Our target OGF is $F(z) = G_{<}(z) + G_{=}(z)$. Solving the system gives:

$$F(z) = \frac{U(z) + G_{=}(z)}{2}$$

Now $G_{=}(z)$ is the OGF for compositions in which the sum of the first two variables equals the sum of the last two.

- The number of compositions with $x_1 + x_2 = n$ is: $[z^n] \frac{1}{(1-z)^2} = n + 1$.
- The number of compositions with $x_3 + x_4 = n$ is the same.
- For the equality $x_1 + x_2 = x_3 + x_4$ to hold, both sums must equal some $n \geq 0$.

$$G_=(z) = \sum_{n=0}^{\infty} (n+1)^2 z^{2n} = \frac{1+z^2}{(1-z^2)^3}$$

Substituting $U(z)$ and $G_=(z)$ into our formula yields:

$$F(z) = \frac{1}{2} \left[\frac{1}{(1-z)^4} + \frac{1+z^2}{(1-z^2)^3} \right] = \frac{1+z+2z^2}{(1-z)^4(1+z)^3}$$

Problem 5

$$\{x_1 \leq x_2, x_2 \geq x_3, x_3 \leq x_4\}$$

Because the inequalities alternate direction, the simple slack-variable chain for $x_1 \leq x_2 \leq x_3 \leq x_4$ no longer applies. Instead, we use inclusion–exclusion. We wish to enforce three conditions

- $C_1 : x_1 \leq x_2$
- $C_2 : x_2 \geq x_3$
- $C_3 : x_3 \leq x_4$

It is easiest to keep the two outer inequalities first, and then subtract the cases where the middle constraint fails. So first consider the set of compositions satisfying both $x_1 \leq x_2$ and $x_3 \leq x_4$. The OGF for this set is easy to obtain because the two pairs are independent.

- For $x_1 \leq x_2$, set $x_2 = x_1 + y_1$. The total contribution is $2x_1 + y_1$, so the OGF is $\frac{1}{(1-z^2)(1-z)}$.
- For $x_3 \leq x_4$, the OGF is identically $\frac{1}{(1-z^2)(1-z)}$.

Since the two pairs are independent, we multiply their OGFs:

$$A(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

Now we find the OGF for the violation $x_1 \leq x_2 < x_3 \leq x_4$ by introducing slack variables

- $x_1 = y_1$
- $x_2 = y_1 + y_2$
- $x_3 = y_1 + y_2 + 1 + y_3$
- $x_4 = y_1 + y_2 + 1 + y_3 + y_4$

Summing the four expressions gives

$$S = 4y_1 + 3y_2 + 2y_3 + y_4 + 2$$

The OGF for this violation is therefore

$$V(z) = \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)}$$

The OGF for the zigzag constraint is the OGF for the two independent pairs minus the OGF for the cases where the peak at x_2 fails:

$$\begin{aligned} F(z) &= \frac{1}{(1-z^2)^2(1-z)^2} - \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)} \\ &= \frac{1}{(1-z)(1-z^2)} \left[\frac{1}{(1-z)(1-z^2)} - \frac{z^2}{(1-z^3)(1-z^4)} \right] \end{aligned}$$

▷ **I.18.** *Odd versus distinct summands.*

This bijection looks clever at first and almost inevitable afterwards. We build a one-to-one correspondence between partitions into odd parts and partitions into distinct parts.

- Odd $\rightarrow$ Distinct
 1. Group equal odd parts together.
 2. If an odd part k appears m times, write m in binary as $m = 2^{a_1} + 2^{a_2} + \dots$ with distinct powers of two.
 3. Replace those m copies of k by the distinct parts $2^{a_1} \cdot k, 2^{a_2} \cdot k, \dots$
- Distinct $\rightarrow$ Odd
 1. Factor each part as $2^a \times \text{odd}$.
 2. Replace it by 2^a copies of that odd factor.
 3. Do this for every part; the result is a partition into odd parts.
 4. Sort the resulting parts to obtain a partition into odd parts.

Here is the map in action:

$$\begin{aligned} 3 + 3 + 3 + 3 + 3 + 1 + 1 &= 5 \times 3 + 2 \times 1 \\ &= (2^2 + 2^0) \times 3 + 2^1 \times 1 \\ &= 4 \times 3 + 1 \times 3 + 2 \times 1 \\ &= 12 + 3 + 2 \end{aligned}$$

$$\begin{aligned} 12 + 3 + 2 &= 4 \times 3 + 3 + 2 \times 1 \\ &= (3 + 3 + 3 + 3) + 3 + (1 + 1) \\ &= 3 + 3 + 3 + 3 + 3 + 1 + 1 \end{aligned}$$

▷ **I.19.** *Euler's pentagonal number theorem.*

To get a usable recurrence for the partition numbers, it is natural to begin with the reciprocal of $P(z)$:

$$A(z) = \prod_{n=1}^{\infty} (1 - z^n)$$

Euler's Pentagonal Number Theorem says that this product collapses to a remarkably sparse series:

$$\begin{aligned} A(z) &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k+1)/2} \\ &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k-1)/2} \\ &= 1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + \dots \end{aligned}$$

The exponents $g_k = \frac{k(3k-1)}{2}$ are the generalized pentagonal numbers. Since $A(z)$ is the reciprocal of $P(z)$, this gives

$$\sum_{n=0}^{\infty} p(n) z^n \cdot \sum_{k=-\infty}^{\infty} (-1)^k z^{g_k} = 1$$

Comparing coefficients of z^n now gives a recurrence that is genuinely useful in computation:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) \dots$$

A straightforward Python implementation follows:

```

1 def compute_partitions(N):
2     P = [0] * (N + 1)
3     P[0] = 1 # Base case: p(0) = 1
4
5     for i in range(1, N + 1):
6         k = 1
7         while True:
8             # Calculate pentagonal numbers for k and -k
9             g_k_pos = k * (3 * k - 1) // 2
10            g_k_neg = k * (3 * k + 1) // 2
11
12            # Determine the sign: +1 for k=1,2, -1 for k=3,4, and
13            # so on.
14            # This is exactly (-1)^(k-1).
15            sign = 1 if k % 2 == 1 else -1
16
17            # Add p(i - g_k_pos)
18            if g_k_pos <= i:
19                P[i] += sign * P[i - g_k_pos]
20            else:
21                break # No more terms fit
22
23            # Add p(i - g_k_neg)
24            if g_k_neg <= i:
25                P[i] += sign * P[i - g_k_neg]
26            else:
27                break
28
29            k += 1
30
31     return P

```

▷ **I.20.** *A digital surprise.*

$$\begin{aligned} \varphi &= (1 - 10^{-1})(1 - 10^{-2})(1 - 10^{-3})(1 - 10^{-4}) \dots \\ &= \prod_{n=1}^{\infty} (1 - z^n) \Big|_{z=.1} \\ &= 1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + z^{22} + z^{26} - \dots \Big|_{z=.1} \\ &= .9 - .1 + .01 + .00001 + .0000001 - .0000000000001 - \dots \end{aligned}$$

The partial sums are

- $\varphi_1 = .99999999999999999999$
- $\varphi_2 = .89999999999999999999$
- $\varphi_3 = .88999999999999999999$
- $\varphi_4 = .89000999999999999999$

- $\varphi_5 = .8900100999999999999999999999$
- $\varphi_6 = .8900100999989999999999999999$
- $\varphi_7 = .8900100999989989999999999999$
- $\varphi_8 = .89001009999899900000000999$

This also explains why only the digits 0, 1, 8, 9 appear. Start from .999... Each new term is either -10^{-m} or $+10^{-m}$ for some generalized pentagonal exponent m , so at each step we modify only one decimal place together with the possible carry or borrow to its left. A subtraction can only change a digit 9 to 8. An addition can create a carry through a block of 9's, turning that block into 0's and eventually producing a 1; if the carry hits an 8, that 8 simply becomes 9. Thus the only possible digit changes are

$$9 \rightarrow 8, \quad 9 \rightarrow 0, \quad 8 \rightarrow 9, \quad 0 \rightarrow 1,$$

and no other digit can ever be created.

▷ **I.21.** *Lattice points.*

We want the number of integer solutions to $\sum_{i=1}^d x_i^2 \leq n^2$. The natural tool here is the theta series. For one coordinate $x_i \in \mathbb{Z}$, the contribution to the squared radius is x_i^2 , so the one-variable generating function is

$$\Theta(z) = \sum_{x \in \mathbb{Z}} z^{x^2} = 1 + 2 \sum_{n=1}^{\infty} z^{n^2}$$

Because the coordinates are independent, the generating function for the total quantity $\sum x_i^2$ is simply the d th power:

$$S(z) = (\Theta(z))^d = \sum_{k=0}^{\infty} r_d(k) z^k$$

Here $r_d(k) = [z^k]S(z)$ counts lattice points whose squared distance from the origin is exactly k . The ball of radius n asks for squared distance at most n^2 , so we need the cumulative sums of the coefficients $r_d(k)$ up to n^2 . Generating functions encode these cumulative sums by multiplying by $\frac{1}{1-z}$:

$$\sum_{k=0}^{\infty} \left(\sum_{j=0}^k r_d(j) \right) z^k = \frac{1}{1-z} S(z) = \frac{1}{1-z} (\Theta(z))^d$$

The coefficient of z^{n^2} is precisely the sum of all $r_d(j)$ with $j \leq n^2$, so

$$N(n, d) = [z^{n^2}] \frac{1}{1-z} (\Theta(z))^d$$

1.4 Words and regular languages

▷ **I.23.** *Alice, Bob, and coding bounds.*

Alice wants to encode an arbitrary binary message into a binary word that avoids the forbidden pattern 11. If messages of length n are encoded into admissible words of length at most L , then we need at least

$$2^n \leq \sum_{j=0}^L C_j,$$

where C_j counts admissible codewords of length j .

By Proposition I.4, the class $\mathcal{C}$ of binary strings avoiding the pattern 11 has generating function

$$C(z) = \frac{1+z}{1-z-z^2}$$

so $C_j = F_{j+2}$, with (F_m) the Fibonacci numbers. Therefore

$$\sum_{j=0}^L C_j = \sum_{j=0}^L F_{j+2} = F_{L+4} - 2.$$

This already gives the exact counting bound

$$2^n \leq F_{L+4} - 2.$$

If one prefers generating functions, the partial sums are encoded by

$$S(z) = \frac{C(z)}{1-z} = \frac{1+z}{(1-z)(1-z-z^2)}.$$

Either way, the asymptotics are governed by the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$, since

$$F_{L+4} \sim \frac{\varphi^{L+4}}{\sqrt{5}}.$$

Thus, as $L \rightarrow \infty$,

$$\sum_{j=0}^L C_j \sim A \cdot \varphi^L$$

for some constant $A > 0$. Taking base-2 logarithms of $2^n \leq \sum_{j=0}^L C_j$ gives

$$n \leq \log_2 \left(\sum_{j=0}^L C_j \right) = L \log_2 \varphi + O(1)$$

and hence

$$L \geq \frac{n}{\log_2 \varphi} + O(1) \approx 1.440420n + O(1)$$

So any such code must incur an asymptotic expansion factor of at least $1.440420 \dots$

▷ **I.27.** *Congruence languages.*

The key observation is that, when a binary digit $b \in \{0, 1\}$ is appended to a word whose current value is congruent to $r \pmod{7}$, the new value is congruent to

$$2r + b \pmod{7}.$$

So it is enough to remember the current remainder modulo 7. This leads to the standard remainder automaton for binary strings congruent to 2 (mod 7):

- **States:** A set $\mathcal{Q} = \{S_0, \dots, S_6\}$, where each state S_r represents the current remainder r modulo 7.
- **Initial State:** S_0 (representing the value 0).
- **Accepting State:** $\overline{\mathcal{Q}} = \{S_2\}$.
- **Alphabet:** $\mathcal{A} = \{0, 1\}$.

Each transition follows the update rule $q_r \circ b = q_{(2r+b) \pmod{7}}$.

Table 1.1: State Transition

Current State (Remainder)	Input 0 ($2r \pmod{7}$)	Input 1 ($2r + 1 \pmod{7}$)
S_0	S_0	S_1
S_1	S_2	S_3
S_2	S_4	S_5
S_3	S_6	S_0
S_4	S_1	S_2
S_5	S_3	S_4
S_6	S_5	S_6

Any binary word is accepted exactly when the automaton finishes in state S_2 , so the language is A-regular.

Nothing here is special to modulus 7 or to base 2. For representations in base b , one replaces the update rule by

$$r \mapsto br + d \pmod{m},$$

where d is the next digit. Thus congruence conditions modulo m are always recognized by a finite automaton, and so are finite boolean combinations of such conditions.

▷ **I.29.** *The forward equations.*

Let $\mathcal{M}_k$ be the set of words that lead from the initial state q_0 to the state q_k . A last-letter decomposition gives

$$\mathcal{M}_k \cong \llbracket k = 0 \rrbracket \cdot \{\epsilon\} + \sum_{j, \alpha: q_j \circ \alpha = q_k} (\mathcal{M}_j \times \{\alpha\})$$

where $\llbracket \cdot \rrbracket$ is the Iverson bracket. Indeed, a word reaching q_k is either the empty word when $k = 0$, or else some word reaching a predecessor state q_j , followed by a final letter α that sends q_j to q_k .

Translating directly to generating functions gives

$$M_k(z) = \delta_{k,0} + z \sum_j M_j(z) T_{j,k}.$$

In vector form, if $\mathbf{M}(z)$ is the row vector $(M_0(z), \dots, M_s(z))$, then

$$\mathbf{M}(z) = \mathbf{e}_0^T + z\mathbf{M}(z)T,$$

where $\mathbf{e}_0^T = (1, 0, \dots, 0)$ is exactly the row vector denoted by u in Proposition I.3. The forward equations count paths from the distinguished initial state q_0 to each state q_k , so the length-0 contribution appears only at $k = 0$. By contrast, the vector v records which states are final, so it belongs to the backward formulation. Once $\mathbf{M}(z)$ is known, the accepted language is recovered only at the end by selecting the final states:

$$L(z) = \mathbf{M}(z)v.$$

▷ **I.33.** *Waiting times in strings.*

Let $L \subseteq \text{SEQ}\{a, b\}$ be a language, and let $\hat{\mathcal{L}}$ denote its prefix language, namely the set of words in L that have no strict prefix belonging to L . For a random infinite binary string with $\mathbb{P}(a) = \mathbb{P}(b) = 1/2$, the quantity $\hat{L}(1/2)$ is the probability that the string starts with a word of $\hat{\mathcal{L}}$, while

$$\frac{1}{2}\hat{L}'(1/2)$$

gives the expected matched length contributed by the first word of the prefix language that is encountered. In the regular case, this expectation is therefore rational. This quantity should be read as a prefix-language statistic: it averages the length of the first matched prefix, with value 0 when no word of $\hat{\mathcal{L}}$ occurs at the beginning.

Let us test these principles on a deliberately tiny prefix language, $\hat{\mathcal{L}} = \{aa, bbb\}$.

1. Prefix property: aa is not a prefix of bbb , and conversely, so the language is prefix-free.
2. Generating function: the OGF is $\hat{L}(z) = z^2 + z^3$.
3. Probability: $\hat{L}(1/2)$ is the probability that a random infinite string starts with a word of $\hat{\mathcal{L}}$: $\hat{L}(1/2) = (1/2)^2 + (1/2)^3 = 1/4 + 1/8 = 3/8$.
4. Expected matched length: $\frac{1}{2}\hat{L}'(1/2)$ gives the expected length contributed by the first matched word: $\hat{L}'(z) = 2z + 3z^2 \implies \hat{L}'(1/2) = 1.75 \implies E = 0.875$.
5. Unpacking the value:
 - Word “aa”: Probability $1/4$ of length 2 $\rightarrow$ Contribution: $2 \times 0.25 = 0.5$.
 - Word “bbb”: Probability $1/8$ of length 3 $\rightarrow$ Contribution: $3 \times 0.125 = 0.375$.
 - No Match: Probability $5/8$ of length 0 $\rightarrow$ Contribution: $0 \times 0.625 = 0$.

▷ **I.34.** *A probabilistic paradox on strings.*

Fix a binary pattern of length k . The generating function $T(z)$ for its first occurrence is:

$$T(z) = \frac{z^k}{z^k + (1 - mz)c(z)} = \frac{z^k}{z^k + (1 - 2z)c(z)}$$

The expected waiting time then follows immediately from the derivative at $z = 1/2$:

$$z \frac{d}{dz} T(z) \Big|_{z=1/2} = 2^k c(1/2)$$

Unlike Note I.33, this is not a prefix-language average with a possible value 0; it is the expected position of the first occurrence of a fixed factor inside the whole infinite sequence. Thus the mean first-occurrence time is governed by the autocorrelation polynomial $c(z)$. Patterns with stronger self-overlap have larger values of $c(1/2)$ and therefore tend to appear later for the first time.

This already leads to a striking comparison. For $p = abb$, the only overlap is the trivial one, so $c(z) = 1$ and the average first-occurrence time is

$$2^3 c(1/2) = 8.$$

For $p' = aaa$, one has the proper self-overlaps of lengths 1 and 2, hence

$$c(z) = 1 + z + z^2,$$

and therefore

$$2^3 c(1/2) = 8 \left(1 + \frac{1}{2} + \frac{1}{4} \right) = 14.$$

So the pattern abb tends to occur sooner than aaa .

At first sight this seems paradoxical. All binary patterns of length 3 have the same expected number of occurrences in a long random word, yet their average first-occurrence times can be quite different. The resolution is that aaa overlaps with itself, so its occurrences tend to come in clusters: once one copy appears, nearby copies become more likely. Those clusters are then separated by longer barren stretches. By contrast, occurrences of abb are more evenly spread. There is therefore no contradiction between equal average counts and unequal average waiting times.

1.5 Tree structures

▷ **I.39.** *Unary–binary trees and Motzkin numbers.*

Unary–binary trees are rooted plane trees in which every node has either 0, 1, or 2 children. Their symbolic specification is therefore

$$\mathcal{M} = \mathcal{Z} \times \text{SEQ}_{\leq 2}(\mathcal{M}),$$

so the characteristic polynomial is

$$\phi(u) = 1 + u + u^2.$$

Consequently, the OGF satisfies

$$M(z) = z(1 + M(z) + M(z)^2),$$

that is,

$$zM(z)^2 - (1 - z)M(z) + z = 0.$$

Solving the quadratic and choosing the branch with $M(0) = 0$ gives

$$M(z) = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z}.$$

Its expansion begins as

$$M(z) = z + z^2 + 2z^3 + 4z^4 + 9z^5 + 21z^6 + 51z^7 + \cdots.$$

The coefficients $M_n = [z^n]M(z)$ are the Motzkin numbers.

To extract them directly, we apply Lagrange inversion:

$$\begin{aligned} M_n &= \frac{1}{n} [u^{n-1}] \phi(u)^n \\ &= \frac{1}{n} [u^{n-1}] (1 + u + u^2)^n \\ &= \frac{1}{n} [u^{n-1}] \sum_j \binom{n}{j} (u + u^2)^j \\ &= \frac{1}{n} \sum_j \binom{n}{j} [u^{n-1}] u^j (1 + u)^j \\ &= \frac{1}{n} \sum_j \binom{n}{j} [u^{n-j-1}] \sum_l \binom{j}{l} u^l \\ &= \frac{1}{n} \sum_j \binom{n}{j} \binom{j}{n-j-1} \\ &= \frac{1}{n} \sum_k \binom{n}{n-k} \binom{n-k}{n-(n-k)-1} \\ &= \frac{1}{n} \sum_k \binom{n}{k} \binom{n-k}{k-1} \end{aligned}$$

which is the explicit form quoted in the book.

▷ **I.41.** *The Lagrangean form of Schröder's GF.*

Only the last equality in the book needs a proof:

$$\frac{(-1)^{n-1}}{n} \sum_k (-2)^k \binom{n}{k+1} \binom{n+k-1}{k} = \frac{1}{n} \sum_{k=0}^{n-2} \binom{2n-k-2}{n-1} \binom{n-2}{k}$$

so it is enough to show

$$(-1)^{n-1} \sum_{k=0}^{n-1} (-2)^k \binom{n}{k+1} \binom{n+k-1}{k} = \sum_{k=0}^{n-2} \binom{2n-k-2}{n-1} \binom{n-2}{k}$$

Let the left-hand side be L and the right-hand side be R .

Re-index the left sum by $j = n - 1 - k$. Using $\binom{n}{k+1} = \binom{n}{n-1-k}$ and $\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$, we get

$$L = (-1)^{n-1} \sum_{j=0}^{n-1} (-2)^{n-1-j} \binom{n}{j} \binom{2n-2-j}{n-1} = 2^{n-1} \sum_{j=0}^{n-1} \left(-\frac{1}{2}\right)^j \binom{n}{j} \binom{2n-2-j}{n-1}$$

Since $\binom{2n-2-j}{n-1} = [z^{n-1}](1+z)^{2n-2-j}$, coefficient extraction yields

$$\begin{aligned} L &= 2^{n-1} \sum_{j=0}^{n-1} \left(-\frac{1}{2}\right)^j \binom{n}{j} [z^{n-1}](1+z)^{2n-2-j} \\ &= 2^{n-1} [z^{n-1}] \sum_{j=0}^{n-1} \left(-\frac{1}{2}\right)^j \binom{n}{j} (1+z)^{2n-2-j} \\ &= 2^{n-1} [z^{n-1}] (1+z)^{2n-2} \sum_{j=0}^n \binom{n}{j} \left(-\frac{1}{2(1+z)}\right)^j \\ &= 2^{n-1} [z^{n-1}] (1+z)^{2n-2} \left(1 - \frac{1}{2(1+z)}\right)^n \\ &= 2^{n-1} [z^{n-1}] (1+z)^{2n-2} \frac{(1+2z)^n}{2^n (1+z)^n} \\ &= \frac{1}{2} [z^{n-1}] (1+z)^{n-2} (1+2z)^n \end{aligned}$$

Here the upper limit has been extended from $n-1$ to n ; the added term has degree $n-2$ inside the coefficient extraction, so it contributes nothing to $[z^{n-1}]$. Expanding $(1+2z)^n$ now gives

$$\begin{aligned} L &= \frac{1}{2} \sum_{j=0}^n \binom{n}{j} 2^j [z^{n-1-j}] (1+z)^{n-2} \\ &= \frac{1}{2} \sum_{j=0}^n \binom{n}{j} 2^j \binom{n-2}{n-1-j}. \end{aligned}$$

Likewise, $\binom{2n-k-2}{n-1} = [z^{n-1}](1+z)^{2n-k-2}$, so

$$\begin{aligned}
R &= \sum_{k=0}^{n-2} \binom{2n-k-2}{n-1} \binom{n-2}{k} \\
&= \sum_{k=0}^{n-2} [z^{n-1}](1+z)^{2n-k-2} \binom{n-2}{k} \\
&= [z^{n-1}] \sum_{k=0}^{n-2} (1+z)^{2n-k-2} \binom{n-2}{k} \\
&= [z^{n-1}](1+z)^{2n-2} \sum_{k=0}^{n-2} \binom{n-2}{k} (1+z)^{-k} \\
&= [z^{n-1}](1+z)^{2n-2} \left(1 + \frac{1}{1+z}\right)^{n-2} \\
&= [z^{n-1}](1+z)^{2n-2} \frac{(2+z)^{n-2}}{(1+z)^{n-2}} \\
&= [z^{n-1}](1+z)^n (2+z)^{n-2} \\
R &= \sum_{k=0}^{n-2} \binom{n-2}{k} 2^{n-2-k} [z^{n-1-k}](1+z)^n \\
&= \sum_{k=0}^{n-2} \binom{n-2}{k} 2^{n-2-k} \binom{n}{k+1}.
\end{aligned}$$

Now set $j = n - 1 - k$. Then

$$R = \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} 2^j \binom{n-2}{n-1-j}.$$

The missing endpoints $j = 0$ and $j = n$ contribute nothing, since

$$\binom{n-2}{n-1} = \binom{n-2}{-1} = 0.$$

Therefore

$$R = \frac{1}{2} \sum_{j=0}^n \binom{n}{j} 2^j \binom{n-2}{n-1-j} = L,$$

which is exactly the last equality claimed in the book.

▷ **I.42.** *Faster determination of Schröder numbers.*

Start from the closed form

$$S(z) = \frac{1}{4}(1+z-\sqrt{1-6z+z^2}).$$

Equivalently,

$$1+z-4S(z) = \sqrt{1-6z+z^2}.$$

Differentiating both sides gives

$$1 - 4S'(z) = \frac{z - 3}{\sqrt{1 - 6z + z^2}}.$$

Let $R(z) = 1 - 6z + z^2$. Multiplying the differentiated relation by $R(z)$ gives

$$R(z)(1 - 4S'(z)) = (z - 3)\sqrt{R(z)}.$$

Now use $\sqrt{R(z)} = 1 + z - 4S(z)$:

$$R(z)(1 - 4S'(z)) = (z - 3)(1 + z - 4S(z)).$$

Expanding gives

$$R(z) - 4R(z)S'(z) = (z - 3)(1 + z) - 4(z - 3)S(z).$$

Since $R(z) - (z - 3)(1 + z) = 4(1 - z)$, this rearranges to

$$R(z)S'(z) + (3 - z)S(z) + z - 1 = 0,$$

that is, the linear differential equation

$$(z^2 - 6z + 1)S'(z) + (3 - z)S(z) + z - 1 = 0.$$

Write

$$S(z) = \sum_{n \geq 1} S_n z^n, \quad S'(z) = \sum_{n \geq 0} (n + 1)S_{n+1} z^n.$$

Substituting into the differential equation and extracting the coefficient of z^n for $n \geq 2$ gives

$$(n - 1)S_{n-1} - 6nS_n + (n + 1)S_{n+1} + 3S_n - S_{n-1} = 0,$$

that is,

$$(n - 2)S_{n-1} - 3(2n - 1)S_n + (n + 1)S_{n+1} = 0.$$

Replacing n by $n + 1$ yields the recurrence in the book:

$$(n + 2)S_{n+2} - 3(2n + 1)S_{n+1} + (n - 1)S_n = 0, \quad n \geq 1.$$

The underlying mechanism is more general. Whenever an algebraic generating function is given in the form

$$p(z)F(z) + q(z) = \sqrt[k]{r(z)},$$

one can differentiate, eliminate the radical, and often obtain a linear differential equation for $F(z)$. Coefficient extraction then turns that differential equation into a recurrence for the coefficients. In the present case, this general holonomic principle leads to the three-term recurrence above.

▷ **I.43.** *Fast determination of the Cayley–Pólya numbers.*

For rooted non-plane trees, Cayley's functional equation is

$$H(z) = z \exp \left(H(z) + \frac{1}{2}H(z^2) + \cdots \right)$$

or, equivalently,

$$H(z) = z \exp \left(\sum_{k \geq 1} \frac{1}{k} H(z^k) \right).$$

Since no closed form is available, the useful point is that logarithmic differentiation still gives an efficient recurrence. Taking the logarithm gives

$$\log H(z) = \log z + \sum_{k \geq 1} \frac{1}{k} H(z^k),$$

and differentiating yields

$$\frac{H'(z)}{H(z)} = \frac{1}{z} + \sum_{k \geq 1} z^{k-1} H'(z^k).$$

Multiplying by $zH(z)$ clears the denominator:

$$zH'(z) = H(z) \left(1 + \sum_{k \geq 1} z^k H'(z^k) \right).$$

Now write $H(z) = \sum_{n \geq 1} H_n z^n$. The inner sum has the divisor form

$$\sum_{k \geq 1} z^k H'(z^k) = \sum_{k \geq 1} \sum_{m \geq 1} m H_m z^{km} = \sum_{\ell \geq 1} \left(\sum_{d|\ell} d H_d \right) z^\ell.$$

Thus, if

$$a_\ell = \sum_{d|\ell} d H_d,$$

then coefficient extraction in

$$zH'(z) = H(z) \left(1 + \sum_{\ell \geq 1} a_\ell z^\ell \right)$$

gives, for $n \geq 2$,

$$(n-1)H_n = \sum_{l=1}^{n-1} H_{n-l} \left(\sum_{d|l} d H_d \right)$$

or, equivalently,

$$H_n = \frac{1}{n-1} \sum_{l=1}^{n-1} H_{n-l} \left(\sum_{d|l} d H_d \right) = \frac{1}{n-1} \sum_{l=1}^{n-1} a_l H_{n-l}.$$

This determines H_n from earlier values only. Computing the divisor sums a_l as the values are needed gives a polynomial-time algorithm for the Cayley–Polya numbers.

To see the recurrence at work, start from the base case $H_1 = 1$, corresponding to the one-node tree.

- To find H_2 , we set $n = 2$. The summation runs for $l = 1$. The only divisor of 1 is 1. $a_1 = \sum_{d|1} d H_d = 1 \cdot H_1 = 1 \implies H_2 = \frac{1}{2-1} (H_{2-1} \cdot a_1) = 1$.

- For H_3 , the summation runs for $l = 1$ and $l = 2$. The divisors of 2 are 1 and 2. $a_2 = \sum_{d|2} dH_d = 1 \cdot H_1 + 2 \cdot H_2 = 3 \implies H_3 = \frac{1}{3-1}(H_{3-1} \cdot a_1 + H_{3-2} \cdot a_2) = 2$.
- For H_4 , the summation runs for $l = 1, 2, 3$. The divisors of 3 are 1 and 3. $a_3 = \sum_{d|3} dH_d = 1 \cdot H_1 + 3 \cdot H_3 = 7 \implies H_4 = \frac{1}{4-1}(H_{4-1} \cdot a_1 + H_{4-2} \cdot a_2 + H_{4-3} \cdot a_3) = 4$.

▷ **I.50.** *Excursions, bridges, and meanders.*

We use the step set $\{-1, +1\}$. A meander is a walk whose partial sums are always non-negative, a bridge is a walk whose final sum is 0, and an excursion is both a meander and a bridge. Thus excursions are exactly Dyck paths.

A meander ending at altitude k can be read as k up-steps that create new floors, with Dyck excursions $\mathcal{D}$ inserted before, between, and after these up-steps:

$$\mathcal{M} = \mathcal{D} + (\mathcal{D} \times \mathcal{Z} \times \mathcal{D}) + (\mathcal{D} \times \mathcal{Z} \times \mathcal{D} \times \mathcal{Z} \times \mathcal{D}) + \cdots = \mathcal{D} \times \sum_{k=0}^{\infty} (\mathcal{Z} \times \mathcal{D})^k$$

Therefore

$$M(z) = \frac{D(z)}{1 - zD(z)}.$$

A bridge is a sequence of primitive excursions. Each primitive excursion leaves 0, avoids returning in the middle, and comes back only at the end.

- **Going Above:** Starts with $+1$, ends with -1 , with a Dyck path in between: $\mathcal{Z} \times \mathcal{D} \times \mathcal{Z}$
- **Going Down:** Starts with -1 , ends with 1 , with a mirrored Dyck path in between: $\mathcal{Z} \times \mathcal{D} \times \mathcal{Z}$

$$\mathcal{B} = \text{SEQ}(\mathcal{Z} \times \mathcal{D} \times \mathcal{Z} + \mathcal{Z} \times \mathcal{D} \times \mathcal{Z}) = \text{SEQ}(2 \times \mathcal{Z}^2 \times \mathcal{D})$$

Thus

$$B(z) = \frac{1}{1 - 2z^2D(z)}.$$

The closed forms of these coefficients are classical. For meanders, let $N(n, k)$ be the number of unrestricted paths of length n ending at altitude k . The reflection principle does the main work: paths that ever hit -1 and end at $k \geq 0$ are in bijection with unrestricted paths ending at $-k-2$. So the number of meanders ending at k is $N(n, k) - N(n, k+2)$, and summing over $k \geq 0$ gives the telescoping identity

$$M_n = \sum_{k \geq 0} [N(n, k) - N(n, k+2)]$$

- n is even ($n = 2m$) $M_n = N(n, 0) = N(2m, 0) = \binom{2m}{\frac{2m+0}{2}} = \binom{2m}{m} = \binom{n}{\lfloor n/2 \rfloor}$
- n is odd ($n = 2m + 1$) $M_n = N(n, 1) = N(2m + 1, 1) = \binom{2m+1}{m+1} = \binom{2m+1}{m} = \binom{n}{\lfloor n/2 \rfloor}$

Hence

$$M_n = \binom{n}{\lfloor n/2 \rfloor}.$$

For bridges, the final altitude is 0, so only even lengths contribute, and an unrestricted choice of which n of the $2n$ steps are up-steps gives

$$B_{2n} = \binom{2n}{n}.$$

Example I.17. *The complexity of boolean functions.*

Let Ω be the set of all boolean functions in m variables. Then

$$\|\Omega\| = 2^{2^m}.$$

As descriptions, use binary expression trees whose internal nodes are labelled by $\vee$ or $\wedge$, and whose external nodes are labelled by one of the $2m$ literals $x_1, \neg x_1, \dots, x_m, \neg x_m$. The size of a description is the number of internal nodes, namely the number of logical connectives.

If a description has size n , its underlying full binary tree has n internal nodes and $n + 1$ external nodes. There are

$$\frac{1}{n+1} \binom{2n}{n}$$

such tree shapes. The internal nodes have 2^n possible connective labels, and the leaves have $(2m)^{n+1}$ possible literal labels. Hence

$$D_n = \frac{1}{n+1} \binom{2n}{n} 2^n (2m)^{n+1}.$$

Using the Catalan estimate

$$\frac{1}{n+1} \binom{2n}{n} = O(4^n n^{-3/2}),$$

we get, for fixed m ,

$$D_n = O(16^n m^n n^{-3/2}).$$

Equivalently, keeping the dependence on m visible,

$$D_n = O(m(16m)^n n^{-3/2}).$$

The counting argument is simple. If

$$\sum_{j=0}^{\nu} D_j < \|\Omega\|,$$

then descriptions of size at most ν cannot represent all boolean functions. If the left-hand side is $o(\|\Omega\|)$, then the fraction of functions of complexity at most ν tends to 0.

Here is the elementary estimate behind the partial sum. If $c > 1$, $\alpha > 0$, and $a_j = c^j j^{-\alpha}$, then

$$\sum_{j=1}^{\nu} a_j = O(c^{\nu} \nu^{-\alpha}).$$

Indeed, split the sum at $\lfloor \nu/2 \rfloor$. For the early part, $j^{-\alpha} \leq 1$ and the geometric sum gives

$$\sum_{j \leq \nu/2} c^j j^{-\alpha} \leq \sum_{j \leq \nu/2} c^j = O(c^{\nu/2}).$$

This is negligible on the target scale, since

$$\frac{c^{\nu/2}}{c^\nu \nu^{-\alpha}} = \frac{\nu^\alpha}{c^{\nu/2}} \longrightarrow 0.$$

For the tail, $j \geq \nu/2$ implies $j^{-\alpha} \leq 2^\alpha \nu^{-\alpha}$, whence

$$\sum_{\nu/2 < j \leq \nu} c^j j^{-\alpha} \leq 2^\alpha \nu^{-\alpha} \sum_{\nu/2 < j \leq \nu} c^j = O(c^\nu \nu^{-\alpha}).$$

Combining both parts proves the estimate.

Apply this with $c = 16m$ and $\alpha = 3/2$. With the explicit factor of m , there are constants C and j_0 such that, for $j \geq j_0$,

$$D_j \leq Cm(16m)^j j^{-3/2}.$$

The finitely many terms with $j < j_0$ only contribute a constant, and the estimate above controls the remaining tail. Therefore

$$\sum_{j=0}^{\nu} D_j = O(m 16^\nu m^\nu \nu^{-3/2})$$

When m is fixed, the leading factor can of course be absorbed into the constant. Now choose

$$\nu(m) = \frac{2^m}{4 + \log_2 m}.$$

Then

$$\nu(m) \log_2(16m) = \frac{2^m}{4 + \log_2 m} (4 + \log_2 m) = 2^m,$$

so

$$16^{\nu(m)} m^{\nu(m)} = 2^{2^m} = \|\Omega\|.$$

Consequently,

$$\sum_{j=0}^{\nu(m)} D_j = O(m \|\Omega\| \nu(m)^{-3/2}) = o(\|\Omega\|).$$

Indeed, the ratio is $O(m \nu(m)^{-3/2})$, which still tends to 0. Thus a fraction tending to 1 of boolean functions in m variables have tree complexity at least

$$\frac{2^m}{4 + \log_2 m}.$$

This lower bound is close to the elementary upper bound. For $m = 1$, the four boolean functions are represented by

$$0 \equiv (x_1 \wedge \neg x_1), \quad 1 \equiv (x_1 \vee \neg x_1), \quad x_1, \quad \neg x_1.$$

For $m > 1$, use the binary decision tree decomposition

$$f(x_1, \dots, x_m) = (\neg x_m \wedge f(x_1, \dots, x_{m-1}, 0)) \vee (x_m \wedge f(x_1, \dots, x_{m-1}, 1)).$$

This reduces an m -variable function to two $(m-1)$ -variable functions and uses three new connectives, so induction gives an upper bound of $2^{m+1} - 3$ connectives. Hence most boolean functions have tree complexity of order $2^m / \log m$, while every one has complexity $O(2^m)$.

1.6 Additional constructions

▷ **I.62.** *Sequences without repeated components.*

A sequence without repeated components is an ordered arrangement of distinct objects drawn from a base class $\mathcal{B}$. The natural route is to build an unordered set of distinct components first, and then order it afterward.

Introduce a marker u for the number of selected components. The bivariate generating function for the power-set construction is

$$A(z, u) = \prod_{\beta \in \mathcal{B}} (1 + uz^{|\beta|}) = \exp \left(\sum_{j \geq 1} (-1)^{j-1} \frac{u^j}{j} B(z^j) \right)$$

where $B(z)$ is the OGF of $\mathcal{B}$. Write

$$A(z, u) = \sum_{k \geq 0} f^{(k)}(z) u^k.$$

Then $f^{(k)}(z)$ counts sets with exactly k distinct components. To turn such sets into sequences, multiply by $k!$, since a set of k distinct objects has $k!$ possible orderings.

Using the Eulerian integral

$$k! = \int_0^\infty e^{-t} t^k \, dt,$$

with t as the integration variable, the OGF of sequences without repeated components is

$$\begin{aligned} \mathcal{S}_{\text{distinct}}(z) &= \sum_{k \geq 0} k! f^{(k)}(z) \\ &= \int_0^\infty e^{-t} \sum_{k \geq 0} f^{(k)}(z) t^k \, dt \\ &= \int_0^\infty A(z, t) e^{-t} \, dt \\ &= \int_0^\infty \exp \left(\sum_{j \geq 1} (-1)^{j-1} \frac{t^j}{j} B(z^j) \right) e^{-t} \, dt. \end{aligned}$$

This is the formula displayed in the book; the integral is formal, serving to implement the factor $k!$ that orders a k -element set.

1.7 Perspective

Chapter 2

LABELLED STRUCTURES AND EXPONENTIAL GENERATING FUNCTIONS

2.1 Labelled classes

2.2 Admissible labelled constructions

2.3 Surjections, set partitions, and words

2.4 Alignments, permutations, and related structures

2.5 Labelled trees, mappings, and graphs

▷ II.19. *Labelled hierarchies.*

The class $\mathcal{L}$ of labelled hierarchies is made of rooted trees whose leaves carry labels, while internal nodes are unlabelled and have outdegree at least 2. The size is the number of labels, so internal nodes do not contribute to size. This gives the labelled specification

$$\mathcal{L} = \mathcal{Z} + \text{SET}_{\geq 2}(\mathcal{L}),$$

and therefore

$$L(z) = z + e^{L(z)} - 1 - L(z).$$

The term $\mathcal{Z}$ is the one-leaf hierarchy. The term $\text{SET}_{\geq 2}(\mathcal{L})$ says that an internal node is just an unordered set of at least two smaller hierarchies.

The book notes that this equation can be solved in terms of the Cayley tree function T , defined by $T(x) = xe^{T(x)}$. Let us derive the displayed form. First rewrite the equation as

$$e^{L(z)} = 2L(z) - z + 1.$$

Now set

$$u = \frac{z-1}{2} - L(z).$$

Then $2L(z) - z + 1 = -2u$ and $-L(z) = u - (z - 1)/2$. Thus

$$1 = (2L(z) - z + 1)e^{-L(z)} = -2ue^{u-(z-1)/2}.$$

Hence

$$ue^u = -\frac{1}{2}e^{(z-1)/2}.$$

Since $W(y)e^{W(y)} = y$ and $T(x) = -W(-x)$, we get

$$u = W\left(-\frac{1}{2}e^{(z-1)/2}\right) = -T\left(\frac{1}{2}e^{z/2-1/2}\right).$$

Substituting back gives

$$L(z) = T\left(\frac{1}{2}e^{z/2-1/2}\right) + \frac{z}{2} - \frac{1}{2}.$$

Expanding the EGF begins as

$$L(z) = z + \frac{z^2}{2!} + 4\frac{z^3}{3!} + 26\frac{z^4}{4!} + 236\frac{z^5}{5!} + \cdots.$$

So the labelled counts start as

$$L_0, L_1, L_2, L_3, L_4, L_5, \dots = 0, 1, 1, 4, 26, 236, \dots$$

II.5.2. Mappings and associated graphs.

Let $\mathcal{T}$ be the class of all Cayley trees, $\mathcal{K}$ the class of connected functional graphs, and $\mathcal{F}$ the class of all mappings. The symbolic decomposition is

$$\mathcal{F} = \text{SET}(\mathcal{K}), \quad \mathcal{K} = \text{CYC}(\mathcal{T}), \quad \mathcal{T} = \mathcal{Z} \star \text{SET}(\mathcal{T}).$$

At the level of EGFs this becomes

$$F(z) = e^{K(z)}, \quad K(z) = \log \frac{1}{1 - T(z)}, \quad T(z) = ze^{T(z)}.$$

Thus

$$F(z) = \frac{1}{1 - T(z)}.$$

This is reassuring, since F_n should count all mappings $[n] \rightarrow [n]$, hence $F_n = n^n$. Here is the Lagrange-inversion check. For $T = z\phi(T)$ with $\phi(w) = e^w$,

$$[z^n]H(T(z)) = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n$$

and taking $H(w) = (1 - w)^{-1}$ gives

$$\begin{aligned}
F_n &= n![z^n]F(z) \\
&= n! \frac{1}{n} [w^{n-1}] \left(\frac{1}{1-w} \right)' (e^w)^n \\
&= (n-1)! [w^{n-1}] \frac{e^{nw}}{(1-w)^2} \\
&= (n-1)! [w^{n-1}] \sum_{k=0}^{\infty} (k+1) w^k \cdot \sum_{j=0}^{\infty} \frac{n^j}{j!} w^j \\
&= (n-1)! \sum_{j=0}^{n-1} ((n-1-j) + 1) \frac{n^j}{j!} \\
&= (n-1)! \sum_{j=0}^{n-1} (n-j) \frac{n^j}{j!} \\
&= (n-1)! \left(\sum_{j=0}^{n-1} n \frac{n^j}{j!} - \sum_{j=0}^{n-1} j \frac{n^j}{j!} \right) \\
&= (n-1)! \left(\sum_{j=0}^{n-1} \frac{n^{j+1}}{j!} - \sum_{j=1}^{n-1} \frac{n^j}{(j-1)!} \right) \\
&= (n-1)! \left(\sum_{j=0}^{n-1} \frac{n^{j+1}}{j!} - \sum_{j=0}^{n-2} \frac{n^{j+1}}{j!} \right) \\
&= (n-1)! \frac{n^n}{(n-1)!} \\
&= n^n
\end{aligned}$$

For connected functional graphs, start from

$$K(z) = \log \frac{1}{1-T(z)} = \sum_{k \geq 1} \frac{T(z)^k}{k}.$$

From Note II.18, for $1 \leq k \leq n$,

$$n![z^n] \frac{T(z)^k}{k!} = \binom{n-1}{k-1} n^{n-k} \implies n![z^n] \frac{T(z)^k}{k} = (k-1)! \binom{n-1}{k-1} n^{n-k} = \frac{(n-1)!}{(n-k)!} n^{n-k}$$

Therefore

$$\begin{aligned}
K_n &= n![z^n]K(z) \\
&= \sum_{k=1}^n \frac{(n-1)!}{(n-k)!} n^{n-k} \\
&= n^{n-1} \sum_{k=1}^n \frac{(n-1)!}{(n-k)! n^{k-1}}.
\end{aligned}$$

The last sum is Ramanujan's Q -function:

$$Q(n) = 1 + \frac{n-1}{n} + \frac{(n-1)(n-2)}{n^2} + \dots + \frac{(n-1)!}{n^{n-1}}.$$

Thus

$$K_n = n^{n-1}Q(n).$$

II.5.2. *Constrained mappings.*

We only look at the first constrained class: mappings without fixed points. In the functional graph decomposition, a fixed point is a 1-cycle, with a rooted tree possibly attached to it. Since

$$\text{CYC}(\mathcal{T}) \rightsquigarrow \log \frac{1}{1-T(z)}$$

and cycles of length 1 contribute exactly $T(z)$, removing them changes the connected component EGF to

$$\log \frac{1}{1-T(z)} - T(z).$$

Thus the EGF of all fixed-point-free mappings is

$$\exp \left(\log \frac{1}{1-T(z)} - T(z) \right) = \frac{e^{-T(z)}}{1-T(z)}.$$

We want to recover the direct count, namely the fact that each element has $n-1$ possible images:

$$n![z^n] \frac{e^{-T(z)}}{1-T(z)} = (n-1)^n$$

by Lagrange inversion. For $n \geq 2$, with $H(w) = \frac{e^{-w}}{1-w}$ and $\phi(w) = e^w$, we obtain

$$\begin{aligned} n![z^n] \frac{e^{-T(z)}}{1-T(z)} &= n! \frac{1}{n} [w^{n-1}] \left(\frac{e^{-w}}{1-w} \right)' (e^w)^n \\ &= (n-1)! [w^{n-1}] \frac{w e^{(n-1)w}}{(1-w)^2} \\ &= (n-1)! [w^{n-2}] \frac{e^{(n-1)w}}{(1-w)^2} \\ &= (n-1)! [w^{n-2}] \sum_{k=0}^{\infty} (k+1) w^k \cdot \sum_{j=0}^{\infty} \frac{(n-1)^j}{j!} w^j \\ &= (n-1)! \sum_{j=0}^{n-2} ((n-2-j)+1) \frac{(n-1)^j}{j!} \\ &= (n-1)! \sum_{j=0}^{n-2} (n-1-j) \frac{(n-1)^j}{j!} \\ &= (n-1)! \left(\sum_{j=0}^{n-2} \frac{(n-1)^{j+1}}{j!} - \sum_{j=1}^{n-2} \frac{(n-1)^j}{(j-1)!} \right) \\ &= (n-1)! \left(\sum_{j=0}^{n-2} \frac{(n-1)^{j+1}}{j!} - \sum_{j=0}^{n-3} \frac{(n-1)^{j+1}}{j!} \right) \\ &= (n-1)! \frac{(n-1)^{n-1}}{(n-2)!} \\ &= (n-1)^n \end{aligned}$$

For $n = 1$ the statement is also true, since there is no fixed-point-free map from a singleton to itself.

II.5.3. Unrooted trees and acyclic graphs.

Let $\mathcal{U}$ be the class of unrooted labelled trees. A rooted labelled tree is just an unrooted labelled tree with one vertex distinguished as the root. Therefore, for $n \geq 1$,

$$T_n = nU_n.$$

This immediately gives Cayley's unrooted-tree count

$$U_n = n^{n-2},$$

but the generating-function form is also useful. Start from

$$T(z) = \sum_{n \geq 0} T_n \frac{z^n}{n!}, \quad U(z) = \sum_{n \geq 0} U_n \frac{z^n}{n!}$$

Differentiate $U(z)$ and multiply by z :

$$z \frac{d}{dz} U(z) = z \frac{d}{dz} \sum_{n \geq 0} U_n \frac{z^n}{n!} = z \sum_{n \geq 1} n U_n \frac{z^{n-1}}{n!} = \sum_{n \geq 1} T_n \frac{z^n}{n!} = T(z)$$

So $\frac{d}{dz} U(z) = \frac{T(z)}{z}$. Since $U(0) = 0$, integrating gives

$$U(z) = \int_0^z \frac{T(w)}{w} dw.$$

Now evaluate the integral using the functional equation for rooted trees:

$$T(z) = ze^{T(z)}$$

Write $t = T(w)$. Then $w = te^{-t}$ and

$$dw = e^{-t}(1-t) dt.$$

As w runs from 0 to z , t runs from 0 to $T(z)$, so

$$U(z) = \int_0^{T(z)} \frac{t}{te^{-t}} e^{-t}(1-t) dt = \int_0^{T(z)} (1-t) dt = T(z) - \frac{T(z)^2}{2}.$$

Thus the EGF of all acyclic labelled graphs, i.e. sets of unrooted labelled trees, is

$$A(z) = e^{U(z)} = \exp\left(T(z) - \frac{T(z)^2}{2}\right).$$

▷ II.22. 2-Regular graphs.

We only prove the equinumerosity statement. Take n labelled lines $L_1, \dots, L_n$ in general position: no two lines are parallel and no three lines meet at one point. For the reverse direction below, we also use the usual generic reading of the setup: apart from the original lines themselves, no line contains three of the intersection points. Now turn a cloud into a graph on vertex set $\{1, \dots, n\}$:

- The line L_i becomes vertex i .
- The chosen intersection point $L_i \cap L_j$ becomes the edge $\{i, j\}$.

This is already reversible: an edge set tells us exactly which intersections to choose.

It remains to check the condition. Let d_i be the number of chosen intersection points lying on L_i ; in the graph this is exactly the degree of vertex i . Since a cloud contains n intersection points and each chosen point lies on two lines,

$$\sum_{i=1}^n d_i = 2n.$$

If the cloud has no three collinear points, then no line L_i contains three chosen points, so $d_i \leq 2$ for every i . The displayed sum then forces $d_i = 2$ for every i , hence the associated graph is 2-regular.

Conversely, if the graph is 2-regular, then $d_i = 2$ for every i . Thus no original line contains three chosen intersection points; by the genericity assumption, this means the chosen intersections form a cloud. Therefore clouds on n labelled lines and 2-regular graphs on n labelled vertices are equinumerous.

▷ **II.23.** *Wright's surgery.*

Let W_k be the EGF of connected graphs of excess k , and introduce an edge marker by

$$w_k(z, y) = y^k W_k(zy).$$

This is just a compact way to say that a graph with n vertices and excess k has $n + k$ edges: each vertex contributes one factor zy , and the extra k edges contribute y^k .

Start with a connected graph of excess $k + 1$. Choose one edge, orient it, and remove it. In a generating function with y marking edges, ∂_y points an edge and removes one factor of y ; in this surgery picture, this is exactly the chosen edge that we delete. The factor 2 chooses its orientation, so the left-hand side is

$$2\partial_y w_{k+1}.$$

After the edge is removed, the total excess drops by 1, from $k + 1$ to k . There are two possibilities.

Case 1: the graph stays connected.

If the removed edge was not a bridge, the remaining graph is connected of excess k , with two ordered distinguished vertices: the former tail and head of the removed directed edge. Choosing an ordered pair of distinct vertices in a labelled graph is represented by

$$z^2 \partial_z^2 w_k.$$

However, not every ordered pair is allowed. If the two chosen vertices are already joined by an edge, adding back the deleted edge would create a multiple edge, which is forbidden for simple graphs. Ordered adjacent pairs are counted by choosing an edge and orienting it, namely $2y\partial_y w_k$. Thus the connected case contributes

$$z^2 \partial_z^2 w_k - 2y\partial_y w_k.$$

Case 2: the graph splits into two components.

If the removed edge was a bridge, the remaining graph has two connected components. Suppose the component containing the tail has excess h ; then the other component has excess $k - h$, because the total remaining excess is k . We must mark one vertex in each component, so the contribution is

$$(z\partial_z w_h)(z\partial_z w_{k-h}).$$

The possible excess values start at -1 , since trees have excess -1 . Hence h ranges from -1 to $k + 1$, which also keeps $k - h \geq -1$.

Putting the connected and disconnected cases together gives Wright's differential recurrence:

$$2\partial_y w_{k+1} = (z^2 \partial_z^2 w_k - 2y \partial_y w_k) + \sum_{h=-1}^{k+1} (z\partial_z w_h) \cdot (z\partial_z w_{k-h})$$

The same recurrence can then be specialized to $W_k(z)$ by setting $y = 1$.

2.6 Additional constructions

▷ **II.28.** *Do two permutations generate the symmetric group?*

The index in the book appears to have a typo here: the relation should involve I_{n+1} , not I_n . We prove

$$n!^2 \pi_n = (n-1)! I_{n+1}$$

where I_m denotes the number of indecomposable permutations of size m .

The first useful observation is that the EGF of pairs of permutations is the same formal series as the ordinary generating function of permutations:

$$\sum_{n \geq 0} n!^2 \frac{z^n}{n!} = \sum_{n \geq 0} n! z^n,$$

so write

$$P(z) = \sum_{n \geq 0} n! z^n.$$

By the exponential formula, the EGF $C(z)$ of connected pairs satisfies

$$C(z) = \log P(z).$$

Since every permutation decomposes uniquely into a sequence of indecomposable blocks, the generating function $I(z)$ of indecomposable permutations satisfies

$$I(z) = \sum_{n \geq 1} I_n z^n, \quad P(z) = \frac{1}{1 - I(z)}, \quad I(z) = 1 - \frac{1}{P(z)}.$$

Now differentiate and extract the consequence:

$$C'(z) = \frac{P'(z)}{P(z)} = P'(z)(1 - I(z)) = P'(z) - P'(z)I(z)$$

Next we extract the coefficient of z^{n-1} from both sides.

Step 1. Extract the left-hand side coefficient.

Since $C(z) = \sum_{n \geq 1} (n!^2 \pi_n) \frac{z^n}{n!}$, its derivative is:

$$C'(z) = \sum_{n \geq 1} (n!^2 \pi_n) \frac{z^{n-1}}{(n-1)!}$$

So the coefficient of z^{n-1} is:

$$[z^{n-1}]C'(z) = \frac{n!^2 \pi_n}{(n-1)!} = n \cdot n! \pi_n$$

Step 2. Extract the right-hand side coefficient.

First simplify the right-hand side. Differentiating $P(z)(1 - I(z)) = 1$ gives

$$P'(z)(1 - I(z)) - P(z)I'(z) = 0 \implies P'(z) - P'(z)I(z) = P(z)I'(z)$$

Therefore

$$[z^{n-1}](P'(z) - P'(z)I(z)) = [z^{n-1}]P(z)I'(z) = \sum_{k=1}^n k I_k(n-k)!$$

Call this last sum S_n . We now show directly that $S_n = I_{n+1}$.

The only recurrence we need comes from $P(z) = 1 + I(z)P(z)$. Extracting $[z^N]$ for $N \geq 1$ gives

$$\sum_{k=1}^N I_k(N-k)! = N! \tag{2.1}$$

Now apply this with $N = n+1$, and separate the last term:

$$\sum_{k=1}^n I_k(n+1-k)! = (n+1)! - I_{n+1}. \tag{2.2}$$

On the other hand, rewrite the same left-hand side by pulling out the factor $n+1-k$:

$$\begin{aligned} \sum_{k=1}^n I_k(n+1-k)! &= \sum_{k=1}^n I_k(n+1-k)(n-k)! \\ &= (n+1) \sum_{k=1}^n I_k(n-k)! - \sum_{k=1}^n k I_k(n-k)! \end{aligned}$$

The first sum is $n!$ by (2.1) with $N = n$, and the second sum is exactly S_n . Hence

$$\sum_{k=1}^n I_k(n+1-k)! = (n+1)! - S_n.$$

Comparing this with (2.2) gives $S_n = I_{n+1}$.

Step 3. Conclude the identity.

Finally, $n \cdot n! \pi_n = S_n$, so

$$n!^2 \pi_n = (n-1)! I_{n+1}.$$

2.7 Perspective

Chapter 3

COMBINATORIAL PARAMETERS AND MULTIVARIATE GENERATING FUNCTIONS

3.1 An introduction to bivariate generating functions (BGFs)

3.2 Bivariate generating functions and probability distributions

3.3 Inherited parameters and ordinary MGFs

▷ **III.7.** *Largest summands in compositions.*

Let M be the largest summand in a random composition. We claim that for any $\epsilon \in (0, 1)$,

$$\mathbb{P}((1 - \epsilon) \log_2 n \leq M \leq (1 + \epsilon) \log_2 n) \rightarrow 1$$

Step 1. Upper bound via the first moment method.

Set $r_+ = \lceil (1 + \epsilon) \log_2 n \rceil$. If $M \geq r_+$, then some summand has size at least r_+ . Let $S = \sum_{k=r_+}^n X_k$, where X_k counts summands of size exactly k . Then S counts summands of size at least r_+ , so

$$\mathbb{P}(M \geq r_+) = \mathbb{P}(S \geq 1) \leq \frac{\mathbb{E}[S]}{1} = \sum_{k=r_+}^n \mathbb{E}[X_k]$$

Use the mean estimate from Proposition III.4 in its explicit form. For $1 \leq k \leq n$,

$$\mathbb{E}[X_k] = \frac{n - k + 3}{2^{k+1}} \leq \frac{n + 2}{2^{k+1}}.$$

For all large enough n , this is at most $\frac{n}{2^k}$, which is more than enough here. Thus

$$\mathbb{E}[S] \leq \sum_{k=r_+}^n \frac{n}{2^k} < \frac{n}{2^{r_+-1}} \leq \frac{2n}{2^{(1+\epsilon)\log_2 n}} = 2n^{-\epsilon} \rightarrow 0$$

$M > (1 + \epsilon) \log_2 n$ implies $M \geq \lceil (1 + \epsilon) \log_2 n \rceil = r_+$, so

$$\mathbb{P}(M > (1 + \epsilon) \log_2 n) \leq \mathbb{P}(M \geq r_+) \rightarrow 0$$

Step 2. Lower bound via the second moment method.

For the lower bound, view a composition of n as $n - 1$ independent gaps, each cut with probability $1/2$. Let $r_- = \lceil (1 - \epsilon) \log_2 n \rceil$. Split the n ones into $m = \lfloor \frac{n}{r_-} \rfloor$ disjoint blocks of length r_- , ignoring the leftovers. Each block has $r_- - 1$ internal gaps, so it contains no internal cut with probability $p = (\frac{1}{2})^{r_- - 1}$.

Define the indicator random variables

$$I_i = \begin{cases} 1 & \text{if block } i \text{ has no internal cuts} \\ 0 & \text{otherwise} \end{cases}$$

for $i = 1, \dots, m$. These indicators depend on disjoint sets of gaps, so they are independent Bernoulli variables with common parameter p . Let $Y = \sum_{i=1}^m I_i$, the number of blocks that survive intact. If $Y \geq 1$, then at least one block is contained in a summand of size at least r_- ; the summand may extend beyond the block, but that only helps. Therefore $\mathbb{P}(M \geq r_-) \geq \mathbb{P}(Y > 0)$.

Now the Paley–Zygmund inequality does exactly the required job:

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]}$$

In our setting, $Y \sim \text{Binomial}(m, p)$.

First compute the expectation:

$$\mathbb{E}[Y] = mp = \left\lfloor \frac{n}{r_-} \right\rfloor \left(\frac{1}{2} \right)^{r_- - 1}$$

Let $a = (1 - \epsilon) \log_2 n$. Since $r_- = \lceil a \rceil$, we have $r_- - 1 < a$. Also $r_- \leq a + 1$, so for all large enough n , $a \geq 1$ and $n \geq 4a$, giving

$$\left\lfloor \frac{n}{r_-} \right\rfloor \geq \frac{n}{r_-} - 1 \geq \frac{n}{a+1} - 1 \geq \frac{n}{2a} - 1 \geq \frac{n}{4a}.$$

Also,

$$\left(\frac{1}{2} \right)^{r_- - 1} > \left(\frac{1}{2} \right)^a = n^{-(1-\epsilon)}.$$

Putting these together,

$$\mathbb{E}[Y] > \frac{n^\epsilon}{4(1-\epsilon)\log_2 n} \rightarrow \infty \quad \text{as } n \rightarrow \infty$$

Next control the second moment. Since Y is binomial,

$$\text{Var}(Y) = mp(1-p) \leq mp = \mathbb{E}[Y]$$

Substitute this into Paley–Zygmund:

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]} = \frac{(\mathbb{E}[Y])^2}{\text{Var}(Y) + (\mathbb{E}[Y])^2} \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y] + (\mathbb{E}[Y])^2} = \frac{1}{1 + \frac{1}{\mathbb{E}[Y]}} \rightarrow 1$$

Finally, $M \geq r_-$ implies $M \geq (1 - \epsilon) \log_2 n$, so

$$\mathbb{P}(M \geq (1 - \epsilon) \log_2 n) \geq \mathbb{P}(M \geq r_-) \geq \mathbb{P}(Y > 0) \rightarrow 1$$

Together with the upper-tail estimate, this proves the claim, since

$$\mathbb{P}(M < (1 - \epsilon) \log_2 n \text{ or } M > (1 + \epsilon) \log_2 n) \rightarrow 0.$$

Step 3. Direct proof of the Paley–Zygmund bound.

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]}$$

Start from $\mathbb{E}[Y]$ and split according to whether $Y > 0$:

$$\mathbb{E}[Y] = \mathbb{E}[Y \cdot \mathbf{1}_{Y>0}] + \mathbb{E}[Y \cdot \mathbf{1}_{Y=0}] = \mathbb{E}[Y \cdot \mathbf{1}_{Y>0}]$$

Now apply Cauchy–Schwarz:

$$\mathbb{E}[Y] \leq \sqrt{\mathbb{E}[Y^2]} \sqrt{\mathbb{E}[\mathbf{1}_{Y>0}]} = \sqrt{\mathbb{E}[Y^2]} \sqrt{\mathbb{P}(Y > 0)}$$

Since everything is nonnegative, we may square both sides:

$$(\mathbb{E}[Y])^2 \leq \mathbb{E}[Y^2] \cdot \mathbb{P}(Y > 0) \implies \mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]}$$

3.4 Inherited parameters and exponential MGFS

▷ III.10. A hundred prisoners II.

The randomisation trick is easiest to see in a toy version with 10 prisoners, where each prisoner may open 5 drawers. Suppose a malicious director puts the cards in one long cycle: physical drawer i contains card $i + 1$, with drawer 10 containing card 1. If the prisoners use the usual cycle-following strategy directly, they all follow this 10-cycle, so the strategy fails.

Before the game, however, the prisoners may secretly randomise the drawer names. For example, let virtual drawer labels correspond to physical drawers as follows:

$$\rho = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 7 & 3 & 9 & 1 & 8 & 2 & 10 & 4 & 6 & 5 \end{pmatrix}.$$

When a prisoner says “open drawer j ”, he actually opens physical drawer $\rho(j)$. The card found there then gives the next virtual drawer label to try.

In this example the effective permutation followed by the prisoners is

$$j \mapsto \sigma(\rho(j)),$$

where σ is the director's physical arrangement. Its cycle decomposition is

$$(1\ 8\ 5\ 9\ 7)(2\ 4)(3\ 10\ 6).$$

All cycles now have length at most 5, so the cycle-following strategy succeeds, even though the director originally chose the worst possible 10-cycle.

The point is not this particular lucky relabelling. If ρ is chosen uniformly at random and kept secret, then for any fixed director's arrangement σ , the product $\sigma \circ \rho$ is uniformly distributed over all permutations. Thus randomising the drawer names reduces the malicious case back to the ordinary random-permutation calculation.

▷ **III.15.** *Distinct component sizes in set partitions.*

The parameter is not the number of components, but the number of *sizes* that occur. So we group components by their size. For each fixed n , there are only two possibilities: use no component of size n , or use a non-empty set of such components and pay one marker u for the fact that size n appears.

For set partitions, components of size n are blocks $\text{SET}_n(\mathcal{Z})$, whose EGF is $z^n/n!$. Thus the contribution of one fixed size n is

$$1 + u \left(\exp \left(\frac{z^n}{n!} \right) - 1 \right) = 1 - u + u \exp \left(\frac{z^n}{n!} \right).$$

Multiplying over all possible block sizes gives the BGF for the number of distinct block sizes:

$$\prod_{n \geq 1} \left(1 - u + u \exp \left(\frac{z^n}{n!} \right) \right).$$

For permutations the same argument applies, with cycles replacing blocks. A cycle of length n has EGF z^n/n , so the fixed-size contribution is

$$1 + u \left(\exp \left(\frac{z^n}{n} \right) - 1 \right) = 1 - u + u \exp \left(\frac{z^n}{n} \right).$$

Hence the BGF for the number of distinct cycle sizes is

$$\prod_{n \geq 1} \left(1 - u + u \exp \left(\frac{z^n}{n} \right) \right).$$

3.5 Recursive parameters

Example III.14. *Leaves and node types in binary trees.*

This is the pruned-binary-tree class from Example I.13: start with an ordinary binary tree and remove the external nodes. The book writes B here, but to avoid confusing it

with ordinary binary trees, we write it as $\bar{B}$. Thus $\bar{B}$ is the class of non-empty plane trees where each node has 0, 1, or 2 ordered children. Its OGF is

$$\bar{B}(z) = C(z) - 1 = \frac{1 - \sqrt{1 - 4z}}{2z} - 1 = \frac{1 - 2z - \sqrt{1 - 4z}}{2z}$$

where $C(z)$ is the usual Catalan generating function, satisfying $C(z) = 1 + zC(z)^2$. Equivalently,

$$\bar{B} = \mathcal{Z} + 2\mathcal{Z}\bar{B} + \mathcal{Z}\bar{B}^2, \quad \bar{B}(z) = z + 2z\bar{B}(z) + z\bar{B}(z)^2.$$

The factor 2 records whether the unique child of a 1-node is on the left or on the right.

The book marks the three node types at once:

$$\bar{B}(z, u_0, u_1, u_2) = zu_0 + 2zu_1\bar{B}(z, u_0, u_1, u_2) + zu_2\bar{B}(z, u_0, u_1, u_2)^2.$$

Lagrange inversion then gives

$$[z^n u_0^{k_0} u_1^{k_1} u_2^{k_2}] \bar{B}(z, u_0, u_1, u_2) = \frac{1}{n} \binom{n}{k_0, k_1, k_2} 2^{k_1},$$

under the natural constraints $k_0 + k_1 + k_2 = n$ and $k_0 = k_2 + 1$. Below we rederive the three first moments directly, one marker at a time.

Leaves

$$\begin{aligned} \bar{B}(z, u) = zu + 2z\bar{B}(z, u) + z\bar{B}(z, u)^2 &\implies \frac{\partial \bar{B}}{\partial u} = z + 2z \frac{\partial \bar{B}}{\partial u} + 2z\bar{B} \frac{\partial \bar{B}}{\partial u} \implies \\ \frac{\partial \bar{B}}{\partial u} = \frac{z}{1 - 2z - 2z\bar{B}} &\implies \left. \frac{\partial \bar{B}}{\partial u} \right|_{u=1} = \frac{z}{1 - 2z - 2z\bar{B}(z)} = \frac{z}{\sqrt{1 - 4z}} \implies \\ [z^n] \left. \frac{\partial \bar{B}}{\partial u} \right|_{u=1} &= [z^{n-1}] (1 - 4z)^{-1/2} = \binom{2n-2}{n-1} = \frac{(2n-2)!}{(n-1)!^2} \end{aligned}$$

1-nodes

$$\begin{aligned} \bar{B}(z, u) = z + 2zu\bar{B}(z, u) + z\bar{B}(z, u)^2 &\implies \frac{\partial \bar{B}}{\partial u} = 2z\bar{B} + 2zu \frac{\partial \bar{B}}{\partial u} + 2z\bar{B} \frac{\partial \bar{B}}{\partial u} \implies \\ \frac{\partial \bar{B}}{\partial u} = \frac{2z\bar{B}}{1 - 2zu - 2z\bar{B}} &\implies \left. \frac{\partial \bar{B}}{\partial u} \right|_{u=1} = \frac{2z\bar{B}(z)}{1 - 2z - 2z\bar{B}(z)} = \frac{1 - 2z}{\sqrt{1 - 4z}} - 1 \implies \\ [z^n] \left. \frac{\partial \bar{B}}{\partial u} \right|_{u=1} &= [z^n] \left((1 - 4z)^{-1/2} - 2 \frac{z}{\sqrt{1 - 4z}} \right) = \binom{2n}{n} - 2 \binom{2n-2}{n-1} \end{aligned}$$

2-nodes

$$\bar{B}(z, u) = z + 2z\bar{B}(z, u) + zu\bar{B}(z, u)^2 \implies \frac{\partial \bar{B}}{\partial u} = 2z \frac{\partial \bar{B}}{\partial u} + z\bar{B}^2 + 2zu\bar{B} \frac{\partial \bar{B}}{\partial u} \implies$$

$$\frac{\partial \bar{B}}{\partial u} = \frac{z\bar{B}^2}{1-2z-2zu\bar{B}} \implies \frac{\partial \bar{B}}{\partial u} \Big|_{u=1} = \frac{z\bar{B}(z)^2}{1-2z-2z\bar{B}(z)} = \frac{z\bar{B}(z)^2}{\sqrt{1-4z}} = \frac{z^3 C(z)^4}{\sqrt{1-4z}}$$

At this point a standard Catalan identity saves a fair amount of algebra. Since $C'(z)$ satisfies $C'(z) = C(z)^2 + 2zC'(z)C'(z)$, we have

$$C'(z) = \frac{C(z)^2}{1-2zC(z)} = \frac{C(z)^2}{\sqrt{1-4z}}$$

Plugging that back in gives:

$$\frac{\partial \bar{B}}{\partial u} \Big|_{u=1} = z^3 C(z)^2 \cdot \frac{C(z)^2}{\sqrt{1-4z}} = z^3 C(z)^2 C'(z) = \frac{z^3}{3} \frac{d}{dz} C(z)^3 = z^3 \frac{d}{dz} F(z)$$

where $F(z) = \frac{1}{3}C(z)^3$. Now:

$$\begin{aligned} [z^n] \frac{\partial \bar{B}}{\partial u} \Big|_{u=1} &= [z^n] z^3 \frac{d}{dz} F(z) \\ &= [z^{n-3}] \frac{d}{dz} F(z) \\ &= (n-2) F_{n-2} \\ &= \frac{n-2}{3} [z^{n-2}] C(z)^3 \\ &= \frac{n-2}{3} \cdot \frac{3}{n+1} \binom{2n-2}{n-2} \\ &= \binom{2n-2}{n-3} \end{aligned}$$

After collecting the three contributions, the total number of leaves, 1-nodes, and 2-nodes over all trees of size n is

$$\binom{2n-2}{n-1} + \binom{2n}{n} - 2 \binom{2n-2}{n-1} + \binom{2n-2}{n-3} = \frac{n}{n+1} \binom{2n}{n}$$

which is exactly $n\bar{B}_n$, as it should be.

Dividing by $\bar{B}_n = \frac{1}{n+1} \binom{2n}{n}$ gives the exact means:

$$\mathbb{E}(\text{leaves}) = \frac{n(n+1)}{2(2n-1)} \sim \frac{n}{4},$$

$$\mathbb{E}(\text{1-nodes}) = \frac{(n+1)(n-1)}{2n-1} \sim \frac{n}{2},$$

$$\mathbb{E}(\text{2-nodes}) = \frac{(n-1)(n-2)}{2(2n-1)} \sim \frac{n}{4}.$$

Thus leaves, left-branching nodes, right-branching nodes, and 2-nodes all have asymptotic proportion $1/4$, since the 1-nodes split equally between left and right.

▷ **III.18.** *Node-degree profile in simple varieties of trees.*

For a simple variety of trees,

$$T(z) = z\phi(T(z)), \quad \phi(w) = \sum_{j \geq 0} \phi_j w^j.$$

To mark nodes with exactly k children, replace the single term $\phi_k T^k$ by $u\phi_k T^k$. Thus

$$T(z, u) = z \left(\phi(T(z, u)) + \phi_k(u-1)T(z, u)^k \right)$$

Let

$$\Omega(z) = \left. \frac{\partial}{\partial u} T(z, u) \right|_{u=1},$$

the cumulative GF for the number of degree- k nodes. Differentiating with respect to u and then setting $u = 1$ gives:

$$\frac{\partial T}{\partial u} = z \left(\phi'(T) \frac{\partial T}{\partial u} + \phi_k T^k + k\phi_k(u-1)T^{k-1} \frac{\partial T}{\partial u} \right) \implies$$

$$\Omega(z) = z \left(\phi'(T(z))\Omega(z) + \phi_k T(z)^k \right) \implies \Omega(z) = z \frac{\phi_k T(z)^k}{1 - z\phi'(T(z))}$$

Since $T(z) = z\phi(T(z))$, differentiating with respect to z yields:

$$T'(z) = \phi(T(z)) + z\phi'(T(z))T'(z) \implies 1 = \frac{\phi(T(z))}{T'(z)} + z\phi'(T(z))$$

Hence,

$$\Omega(z) = \phi_k z T(z)^k \cdot \frac{T'(z)}{\phi(T(z))} = \phi_k z T(z)^k \cdot \frac{T'(z)}{T(z)/z} = \phi_k z^2 T(z)^{k-1} T'(z)$$

For Cayley trees, $\phi(w) = e^w$, so $\phi_k = 1/k!$ and $T(z) = ze^{T(z)}$. Then

$$\begin{aligned} n![z^n]\Omega(z) &= n![z^n] \frac{1}{k!} z^2 \frac{d}{dz} \frac{T(z)^k}{k} \\ &= \frac{n!}{k \cdot k!} [z^{n-2}] \frac{d}{dz} T(z)^k \\ &= \frac{n!}{k \cdot k!} (n-1) [z^{n-1}] T(z)^k \\ &= \frac{n(n-1)}{k} \cdot (n-1)! [z^{n-1}] \frac{T(z)^k}{k!} \\ &= \frac{n(n-1)}{k} \binom{n-2}{k-1} (n-1)^{n-1-k} \\ &= (n-k) \binom{n}{k} (n-1)^{n-1-k} \end{aligned}$$

This is the total number of degree- k nodes over all rooted labelled trees of size n . Since there are n^{n-1} such trees, the mean is

$$\frac{(n-k) \binom{n}{k} (n-1)^{n-1-k}}{n^{n-1}} \sim \frac{ne^{-1}}{k!}$$

for fixed k , matching the statement in Note III.17.

Example III.15. *Path length in trees.*

For general Catalan trees, $G = \mathcal{Z} \times \text{SEQ}(G)$, hence

$$G(z) = \frac{z}{1 - G(z)}.$$

Let $\lambda(\tau)$ be the path length, the sum of distances of all nodes to the root. If τ has root subtrees v , then

$$\lambda(\tau) = \sum_{v \prec \tau} (\lambda(v) + |v|).$$

The extra $|v|$ appears because every node in a root subtree is one level farther from the root of τ than from the root of v .

Write $G(z, u)$ for the BGF where u marks path length. Marking $\lambda(v) + |v|$ instead of just $\lambda(v)$ corresponds to the substitution $z \mapsto zu$, so the root decomposition gives

$$G(z, u) = \frac{z}{1 - G(zu, u)}.$$

Now set

$$\Omega(z) = \left. \frac{\partial}{\partial u} G(z, u) \right|_{u=1}.$$

Differentiate the functional equation with respect to u :

$$G_u(z, u) = \frac{z}{(1 - G(zu, u))^2} (zG_z(zu, u) + G_u(zu, u)).$$

Setting $u = 1$ gives the linear equation

$$\Omega(z) = \frac{z}{(1 - G(z))^2} (zG'(z) + \Omega(z)).$$

Solving for $\Omega(z)$ yields

$$\Omega(z) = \frac{z^2 G'(z)}{(1 - G(z))^2 - z}.$$

It remains to simplify this expression:

$$z^2 \frac{G'(z)}{(1 - G(z))^2 - z} = \frac{z}{2(1 - 4z)} - \frac{z}{2\sqrt{1 - 4z}}$$

The clean route is to rewrite everything in terms of G :

$$G = \frac{z}{1 - G} \implies z = G(1 - G) = G - G^2$$

$$z = G - G^2 \implies 1 = G' - 2GG' \implies G' = \frac{1}{1 - 2G}$$

Hence,

$$\begin{aligned}
z^2 \frac{G'(z)}{(1 - G(z))^2 - z} &= (G(1 - G))^2 \frac{1/(1 - 2G)}{1 - 2G + G^2 - (G - G^2)} \\
&= \frac{G^2(1 - G)^2}{(1 - 2G)(1 - 3G + 2G^2)} \\
&= \frac{G^2(1 - G)}{(1 - 2G)^2} \\
&= \frac{(G - G^2) \cdot G}{1 - 4(G - G^2)} \\
&= \frac{zG}{1 - 4z} \\
&= \frac{z}{1 - 4z} \cdot \frac{1 - \sqrt{1 - 4z}}{2} \\
&= \frac{z}{2} \left(\frac{1}{1 - 4z} - \frac{1}{\sqrt{1 - 4z}} \right)
\end{aligned}$$

▷ **III.20.** *Path length in simple varieties of trees.*

The same path-length recursion used for Catalan trees works for any simple variety: when a subtree is attached below the root, all its nodes move one level down, so its contribution changes from λ to $\lambda + |\cdot|$. This is again the substitution $z \mapsto zu$ inside the recursive call. Thus we start from

$$T(z) = z\phi(T(z)) \text{ where } T(z) = T(z, 1); \quad T(z, u) = z\phi(T(zu, u))$$

Let

$$\Omega(z) = T_u(z, 1)$$

be the cumulative GF of path length. Differentiate the second equation with respect to u :

$$\frac{\partial}{\partial u} T(z, u) = z\phi'(T(zu, u)) \cdot \frac{d}{du} T(zu, u) = z\phi'(T(zu, u)) [T_z(zu, u)z + T_u(zu, u)]$$

At $u = 1$, this becomes

$$T_u(z, 1) = z\phi'(T(z)) [zT'(z) + T_u(z, 1)] \implies T_u(z, 1) = \frac{z^2\phi'(T(z))T'(z)}{1 - z\phi'(T(z))} \quad (3.1)$$

Differentiating the first equation with respect to z gives:

$$T'(z) = \phi(T(z)) + z\phi'(T(z))T'(z) \implies \frac{1}{1 - z\phi'(T(z))} = \frac{T'(z)}{\phi(T(z))}$$

Insert this into (3.1):

$$\Omega(z) = T_u(z, 1) = z^2\phi'(T(z))T'(z) \cdot \frac{T'(z)}{\phi(T(z))} = \frac{\phi'(T(z))}{\phi(T(z))} (zT'(z))^2$$

3.6 Complete generating functions and discrete models

Example III.16. *Words and records.*

After setting all letter markers to 1, the cumulative GF for records is

$$\Omega(z) = \frac{z}{1-rz} \sum_{j=1}^r \frac{1}{1-(j-1)z}.$$

We only extract its coefficient and divide by the total number r^n of words of length n .

$$\begin{aligned} [z^n] \frac{z}{1-rz} \sum_{j=1}^r \frac{1}{1-(j-1)z} &= [z^n] \frac{z}{1-rz} \sum_{k=0}^{r-1} \frac{1}{1-kz} \\ &= [z^n] \sum_{k=0}^{r-1} \frac{z}{(1-rz)(1-kz)} \\ &= [z^n] \sum_{k=0}^{r-1} \frac{1}{r-k} \left(\frac{rz}{1-rz} - \frac{kz}{1-kz} \right) \\ &= \sum_{k=0}^{r-1} \frac{1}{r-k} \left([z^n] \frac{rz}{1-rz} - [z^n] \frac{kz}{1-kz} \right) \\ &= \sum_{k=0}^{r-1} \frac{r^n - k^n}{r-k} \end{aligned}$$

$$\mathbb{E}_{\mathcal{W}_n}(\# \text{records}) = \frac{1}{r^n} [z^n] \Omega(z) = \sum_{k=0}^{r-1} \frac{1 - (k/r)^n}{r-k} = \sum_{k=0}^{r-1} \frac{1}{r-k} - \sum_{k=0}^{r-1} \frac{(k/r)^n}{r-k} = H_r - \sum_{k=1}^{r-1} \frac{(k/r)^n}{r-k}$$

Example III.19. *Rises in Bernoulli trials: Simon Newcomb's problem.*

Let $z_j = zu_j$, where u_j marks occurrences of the letter a_j . Let v mark weak rises: a position followed by a letter that is at least as large as the current one. Write W_j for non-empty words that start with a_j , and write

$$W = 1 + W_1 + \cdots + W_r$$

for all words, including the empty word.

Step 1. Pass from W_j to X_j .

Start from the recurrence:

$$W_j = z_j + z_j \sum_{k=1}^{j-1} W_k + vz_j \sum_{k=j}^r W_k, \quad (3.2)$$

The first term is the one-letter word a_j . If the next letter is smaller than a_j , there is no weak rise, giving the middle sum. If the next letter is at least a_j , we get one weak rise,

giving the factor v in the last sum. It is convenient to pass to X_j :

$$X_j = 1 + \sum_{k=1}^{j-1} W_k + v \sum_{k=j}^r W_k = 1 + \sum_{k=1}^{j-1} z_k X_k + v \sum_{k=j}^r z_k X_k. \quad (3.3)$$

Now take the forward difference:

$$X_{j+1} - X_j = W_j - vW_j = z_j X_j (1 - v),$$

which gives the first-order recurrence:

$$X_{j+1} = X_j (1 + z_j (1 - v)). \quad (3.4)$$

Step 2. Solve the first-order recurrence.

Unwinding (3.4) shows that every X_j is determined by X_1 :

$$X_j = X_1 \prod_{k=1}^{j-1} (1 + (1 - v)z_k). \quad (3.5)$$

Step 3. Determine X_1 .

To determine X_1 , look at the boundary case $j = 1$:

$$W_1 = z_1 + vz_1 \sum_{k=1}^r W_k \implies z_1 X_1 = z_1 + vz_1 \sum_{k=1}^r z_k X_k.$$

Canceling z_1 gives:

$$X_1 = 1 + v \sum_{k=1}^r z_k X_k.$$

To evaluate the remaining sum, introduce the formal boundary term X_{r+1} by setting $j = r + 1$ in (3.3). Then the second sum disappears, leaving

$$X_{r+1} = 1 + \sum_{k=1}^r z_k X_k \implies \sum_{k=1}^r z_k X_k = X_{r+1} - 1.$$

Substituting back gives:

$$X_1 = 1 + v(X_{r+1} - 1).$$

On the other hand, (3.5) at $j = r + 1$ gives $X_{r+1} = X_1 P$, where $P = \prod_{j=1}^r (1 + (1 - v)z_j)$. Plugging this into the equation for X_1 ,

$$X_1 = 1 + v(X_1 P - 1),$$

and solving for X_1 gives:

$$X_1 = \frac{1 - v}{1 - vP}. \quad (3.6)$$

Step 4. Recover the total generating function.

Finally, the total generating function is:

$$W = 1 + \sum_{k=1}^r W_k = 1 + \sum_{k=1}^r z_k X_k = X_{r+1} = \frac{P(1 - v)}{1 - vP}. \quad (3.7)$$

This is the same as the book's form

$$W = \frac{v-1}{v-P^{-1}},$$

after multiplying numerator and denominator by P .

3.7 Additional constructions

Example III.22. *Constrained integer compositions and “slicing”.*

This example concerns two independent regions, $\mathcal{R}_1$ and $\mathcal{R}_2$. They illustrate the same slicing philosophy, but the functional equations and resulting series are derived separately.

Region $\mathcal{R}_1$

Step 1. Derive the functional equation for $\mathcal{R}_1$.

$$\mathcal{R}_1 = \{(x, y) | 1 \leq x \leq y\}$$

To obtain the functional equation, note that

$$\begin{aligned} f(z) &= z + z^2 + \cdots = \frac{z}{1-z} \\ \mathcal{L}[F(z, u)] &= \mathcal{L}\left[\sum_j f^{(j)}(z)u^j\right] \\ &= \sum_j f^{(j)}(z)\mathcal{L}[u^j] \\ &= \sum_j f^{(j)}(z)\frac{u^j}{1-u} \\ &= \frac{1}{1-u}F(z, u) \end{aligned}$$

Therefore,

$$F(z, u) = f(zu) + (\mathcal{L}[F(z, u)])_{u \mapsto zu} = \frac{zu}{1-zu} + \frac{1}{1-zu}F(z, zu) \quad (3.8)$$

Step 2. Check the closed form for $\mathcal{R}_1$.

Now let us check that the two advertised forms of $F(z, u)$ really agree:

$$F(z, u) = \frac{zu}{1-zu} + \frac{z^2u}{(1-zu)(1-z^2u)} + \frac{z^3u}{(1-zu)(1-z^2u)(1-z^3u)} + \cdots \quad (3.9)$$

$$F(z, u) = \frac{zu}{1-z} + \frac{z^2u^2}{(1-z)(1-z^2)} + \frac{z^3u^3}{(1-z)(1-z^2)(1-z^3)} + \cdots \quad (3.10)$$

(3.9) comes from repeated substitution. One could verify (3.10) by series expansion, but there is a cleaner algebraic route. Assume $F(z, u) = \sum_{k \geq 1} A_k(z)u^k$ and substitute into the functional equation (3.8):

$$(1-zu)F(z, u) = zu + F(z, zu)$$

Expanding both sides gives:

$$\sum_{k \geq 1} A_k(z) u^k - zu \left(\sum_{k \geq 1} A_k(z) u^k \right) = zu + \sum_{k \geq 1} A_k(z) (zu)^k$$

Now collect powers of u :

$$\sum_{k \geq 1} A_k(z) u^k - \sum_{k \geq 1} A_k(z) zu^{k+1} = zu + \sum_{k \geq 1} A_k(z) z^k u^k$$

Extract the coefficients of u^k (for $k > 1$):

$$A_k(z) - zA_{k-1}(z) = z^k A_k(z) \implies A_k(z) = \frac{z}{1-z^k} A_{k-1}(z) \quad (3.11)$$

Starting from $A_1(z) = z + z^2 + \dots = \frac{z}{1-z}$, the recurrence telescopes to

$$A_k(z) = \frac{z}{1-z^k} \frac{z}{1-z^{k-1}} \cdots \frac{z}{1-z^2} \frac{z}{1-z} = \frac{z^k}{(1-z)(1-z^2) \cdots (1-z^k)}$$

So the k th term of (3.10) is exactly $A_k(z) u^k$.

Step 3. Compare the two combinatorial interpretations.

There is also a nice combinatorial punchline: the k th term of (3.9) counts partitions with exactly k summands, whereas the k th term of (3.10) counts partitions whose largest summand is exactly k . The case $k = 3$ makes the contrast visible.

The 3rd term in (3.9):

$$\frac{z^3 u}{(1-zu)(1-z^2 u)(1-z^3 u)} = zz(zu) \sum_j (zu)^j \sum_k z^k (zu)^k \sum_l z^l z^l (zu)^l$$

- $z^1 \cdot z^1 \cdot (zu)^1$ places 1 cell on all three columns
- $z^0 \cdot z^0 \cdot (zu)^j$ places $j \geq 0$ cells on the rightmost column
- $z^0 \cdot z^k \cdot (zu)^k$ places $k \geq 0$ cells on the rightmost two columns
- $z^l \cdot z^l \cdot (zu)^l$ places $l \geq 0$ cells on all three columns

This lifting mechanism is what keeps the summands weakly increasing.

The 3rd term in (3.10):

$$\frac{z^3 u^3}{(1-z)(1-z^2)(1-z^3)}$$

- $z^3 u^3$ ensures there is at least one part of size 3 (which also marks the last column height 3).
- $\frac{1}{(1-z)(1-z^2)(1-z^3)}$ is the standard generating function for partitions using parts of size $\{1, 2, 3\}$.

We now leave $\mathcal{R}_1$ entirely and start a separate analysis for $\mathcal{R}_2$.

Region $\mathcal{R}_2$

Step 1. Derive the functional equation for $\mathcal{R}_2$.

$$\mathcal{R}_2 = \{(x, y) | 1 \leq y \leq 2x\}$$

For this second region, the slicing operator changes, so we begin again:

$$\begin{aligned} f(z) &= z \\ \mathcal{L}[F(z, u)] &= \sum_j f^{(j)}(z) \mathcal{L}[u^j] \\ &= \sum_j f^{(j)}(z) u \frac{1 - u^{2j}}{1 - u} \\ &= \frac{u}{1 - u} \sum_j f^{(j)}(z) (1 - u^{2j}) \\ &= \frac{u}{1 - u} \left(\sum_j f^{(j)}(z) - \sum_j f^{(j)}(z) (u^2)^j \right) \\ &= \frac{u}{1 - u} (F(z, 1) - F(z, u^2)) \end{aligned}$$

Hence,

$$F(z, u) = f(zu) + (\mathcal{L}[F(z, u)])_{u \mapsto zu} = zu + \frac{zu}{1 - zu} (F(z, 1) - F(z, z^2u^2)) \quad (3.12)$$

Repeated substitution gives:

$$F(z, u) = \alpha(u) - \frac{zu}{1 - zu} \alpha(z^2u^2) + \frac{zu}{1 - zu} \frac{z^3u^2}{1 - z^3u^2} \alpha(z^6u^4) - \dots$$

where

$$\alpha(u) = zu + \frac{zu}{1 - zu} F(z, 1), \quad \beta(u) = -\frac{zu}{1 - zu}, \quad \sigma(u) = z^2u^2$$

Step 2. Solve for $F(z, 1)$.

Set $u = 1$ and write $X = F(z, 1)$:

$$\begin{aligned} X &= \alpha(1) - \frac{z}{1 - z} \alpha(z^2) + \frac{z}{1 - z} \frac{z^3}{1 - z^3} \alpha(z^6) - \dots \\ &= \left[z + \frac{z}{1 - z} X \right] - \frac{z}{1 - z} \left[z^3 + \frac{z^3}{1 - z^3} X \right] + \frac{z}{1 - z} \frac{z^3}{1 - z^3} \left[z^7 + \frac{z^7}{1 - z^7} X \right] - \dots \end{aligned}$$

This is a linear equation in X :

$$X = A(z) + XB(z). \quad (3.13)$$

From here, separate the constant contribution from the coefficient of X . This gives

$$\begin{aligned}
A(z) &= z - \frac{z}{1-z} z^3 + \frac{z}{1-z} \frac{z^3}{1-z^3} z^7 - \dots \\
&= z - \frac{z^{1+3}}{1-z} + \frac{z^{1+3+7}}{(1-z)(1-z^3)} - \dots \\
&= \sum_{j \geq 1} (-1)^{j-1} z^{2^{j+1}-j-2} / \left((1-z)(1-z^3) \dots (1-z^{2^{j-1}-1}) \right) \\
&= \sum_{j \geq 1} (-1)^{j-1} z^{2^{j+1}-j-2} / Q_{j-1}(z)
\end{aligned}$$

The exponent is:

$$\begin{aligned}
1 + 3 + 7 + \dots + (2^j - 1) &= (2 - 1) + (2^2 - 1) + (2^3 - 1) + \dots + (2^j - 1) \\
&= 2 + 2^2 + 2^3 + \dots + 2^j - j \\
&= \frac{2 - 2^{j+1}}{1 - 2} - j \\
&= 2^{j+1} - j - 2
\end{aligned}$$

$$\begin{aligned}
1 - B(z) &= 1 - \frac{z}{1-z} + \frac{z}{1-z} \frac{z^3}{1-z^3} - \frac{z}{1-z} \frac{z^3}{1-z^3} \frac{z^7}{1-z^7} + \dots \\
&= 1 - \frac{z}{1-z} + \frac{z^{1+3}}{(1-z)(1-z^3)} - \frac{z^{1+3+7}}{(1-z)(1-z^3)(1-z^7)} + \dots \\
&= \sum_{j \geq 0} (-1)^j z^{2^{j+1}-j-2} / \left((1-z)(1-z^3) \dots (1-z^{2^j-1}) \right) \\
&= \sum_{j \geq 0} (-1)^j z^{2^{j+1}-j-2} / Q_j(z)
\end{aligned}$$

Finally,

$$F(z, 1) = X = \frac{\sum_{j \geq 1} (-1)^{j-1} z^{2^{j+1}-j-2} / Q_{j-1}(z)}{\sum_{j \geq 0} (-1)^j z^{2^{j+1}-j-2} / Q_j(z)} \quad (3.14)$$

where

$$Q_j(z) = \prod_{m=1}^j (1 - z^{2^m-1}) = (1-z)(1-z^3)(1-z^7) \dots (1-z^{2^j-1}), \quad Q_0(z) = 1.$$

This conclusion belongs only to $\mathcal{R}_2$; it does not depend on the earlier computation for $\mathcal{R}_1$.

The sequence begins 1, 1, 2, 3, 5, 9, 16, 28, 50 and is OEIS A002572. One interpretation is: the coefficient indexed by m counts partitions of 1 into exactly m summands chosen from

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$$

Here the summand 1 should not be included; otherwise the first terms would be shifted by the trivial partition $1 = 1$. For example, there is 1 way with 2 summands, 1 way with

3 summands, 2 ways with 4 summands, 3 ways with 5 summands, and 5 ways with 6 summands:

$$1 = \frac{1}{2} + \frac{1}{2}$$

$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{4}$$

$$1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$$

$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8}$$

$$1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8}$$

$$1 = \frac{1}{2} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$$

$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16}$$

$$1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$$

$$1 = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16}$$

$$1 = \frac{1}{2} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16}$$

$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16}$$

$$1 = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{32}$$

▷ **III.32.** *Carlitz compositions I.*

To obtain the functional equation, note that

$$k(z) = z + z^2 + \cdots = \frac{z}{1-z}$$

$$\mathcal{L}[K(z, u)] = \sum_j k^{(j)}(z) \mathcal{L}[u^j]$$

$$= \sum_j k^{(j)}(z) \left(\frac{u}{1-u} - u^j \right)$$

$$= \frac{u}{1-u} \sum_j k^{(j)}(z) - \sum_j k^{(j)}(z) u^j$$

$$= \frac{u}{1-u} K(z, 1) - K(z, u)$$

Therefore,

$$K(z, u) = k(zu) + (\mathcal{L}[K(z, u)])_{u \rightarrow zu} = \frac{zu}{1-zu} + \frac{zu}{1-zu} K(z, 1) - K(z, zu)$$

To solve for $K(z, 1)$, write $X = K(z, 1)$:

$$K(z, u) = \frac{zu}{1 - zu}(1 + X) - K(z, zu)$$

One iteration gives:

$$K(z, u) = \frac{zu}{1 - zu}(1 + X) - \left(\frac{z^2u}{1 - z^2u}(1 + X) - K(z, z^2u) \right)$$

Iterating all the way gives:

$$\begin{aligned} K(z, u) &= \frac{zu}{1 - zu}(1 + X) - \frac{z^2u}{1 - z^2u}(1 + X) + \frac{z^3u}{1 - z^3u}(1 + X) - \dots \\ &= (1 + X) \sum_{j \geq 1} (-1)^{j-1} \frac{z^j u}{1 - z^j u} \end{aligned}$$

Now set $u = 1$:

$$X = (1 + X) \sum_{j \geq 1} \frac{(-1)^{j-1} z^j}{1 - z^j}$$

If we call the sum S , then

$$X = (1 + X)S \implies 1 + X = \frac{1}{1 - S} = \left(1 - \sum_{j \geq 1} \frac{(-1)^{j-1} z^j}{1 - z^j} \right)^{-1} = \left(1 + \sum_{j \geq 1} \frac{(-z)^j}{1 - z^j} \right)^{-1}$$

The small bookkeeping point is that this final expression equals $1 + X$, not X itself. In other words, it is the GF where the empty composition is included. For the non-empty class tracked by $K(z, u)$, the answer is instead

$$K(z, 1) = X = \left(1 + \sum_{j \geq 1} \frac{(-z)^j}{1 - z^j} \right)^{-1} - 1.$$

This also explains the constant term in the book's displayed series

$$1 + z + z^2 + 3z^3 + 4z^4 + \dots :$$

that series is $1 + K(z, 1)$.

Example III.23. *Local order patterns in permutations.*

Step 1. Solve the differential equation.

For non-empty increasing binary trees, the boxed-product specification gives

$$\frac{d}{dz} \hat{I}(z, \mathbf{u}) = u_0 + (u_1 + u'_1) \hat{I}(z, \mathbf{u}) + u_2 \hat{I}(z, \mathbf{u})^2, \quad \hat{I}(0, \mathbf{u}) = 0.$$

For brevity, let $\hat{I}(z, \mathbf{u})$ be abbreviated to I , and set

$$v_1 = \frac{u_1 + u'_1}{2}, \quad \delta^2 = u_0 u_2 - v_1^2.$$

Then the differential equation becomes

$$\frac{dI}{dz} = u_0 + (u_1 + u'_1)I + u_2 I^2 = u_0 + 2v_1 I + u_2 I^2$$

Separating variables yields:

$$\frac{dI}{u_0 + 2v_1 I + u_2 I^2} = dz$$

Integrate this from the initial condition $I(0) = 0$ up to a general z :

$$\int_{\xi=I(0)}^{\xi=I(z)} \frac{d\xi}{u_0 + 2v_1 \xi + u_2 \xi^2} = \int_{\zeta=0}^{\zeta=z} d\zeta$$

Using $I(0) = 0$ and writing $I(z)$ simply as I ,

$$\int_0^I \frac{d\xi}{u_0 + 2v_1 \xi + u_2 \xi^2} = \int_0^z d\zeta = z$$

Complete the square in the denominator:

$$u_0 + 2v_1 \xi + u_2 \xi^2 = u_2 \left[\left(\xi + \frac{v_1}{u_2} \right)^2 + \frac{\delta^2}{u_2^2} \right]$$

which turns the integral into

$$\int_0^I \frac{d\xi}{u_2 \left[\left(\xi + \frac{v_1}{u_2} \right)^2 + \left(\frac{\delta}{u_2} \right)^2 \right]} = z$$

Using the standard integral $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan \frac{x}{a}$:

$$z = \frac{1}{u_2} \frac{1}{\frac{\delta}{u_2}} \arctan \frac{\xi + \frac{v_1}{u_2}}{\frac{\delta}{u_2}} \Big|_{\xi=0}^{\xi=I} = \frac{1}{\delta} \left[\arctan \left(\frac{u_2 I + v_1}{\delta} \right) - \arctan \left(\frac{v_1}{\delta} \right) \right]$$

Now rearrange and take tangents:

$$\frac{u_2 I + v_1}{\delta} = \tan \left[\arctan \left(\frac{v_1}{\delta} \right) + z\delta \right] = \frac{\frac{v_1}{\delta} + \tan(z\delta)}{1 - \frac{v_1}{\delta} \tan(z\delta)} = \frac{v_1 + \delta \tan(z\delta)}{\delta - v_1 \tan(z\delta)}$$

Solving for I yields:

$$I = \frac{\delta}{u_2} \frac{v_1 + \delta \tan(z\delta)}{\delta - v_1 \tan(z\delta)} - \frac{v_1}{u_2} \tag{3.15}$$

Step 2. Expand the closed form.

Now expand I as a power series. Start from the closed form (3.15):

The following algebra temporarily divides by v_1 ; the final expansion is a polynomial identity, so this is just a convenient way to organize the terms.

$$\begin{aligned}
I &= \frac{v_1}{u_2} \frac{1 + \frac{\delta}{v_1} \tan(z\delta)}{1 - \frac{v_1}{\delta} \tan(z\delta)} - \frac{v_1}{u_2} \\
&= \frac{v_1}{u_2} \left[1 + \frac{\delta}{v_1} \tan(z\delta) \right] \left[1 + \frac{v_1}{\delta} \tan(z\delta) + \left(\frac{v_1}{\delta} \tan(z\delta) \right)^2 + \left(\frac{v_1}{\delta} \tan(z\delta) \right)^3 + \dots \right] - \frac{v_1}{u_2} \\
&= \frac{v_1}{u_2} \left[1 + \frac{\delta}{v_1} \left(z\delta + \frac{(z\delta)^3}{3} + \dots \right) \right] \left[1 + \frac{v_1}{\delta} \left(z\delta + \frac{(z\delta)^3}{3} \right) + \left(\frac{v_1}{\delta} (z\delta) \right)^2 + \left(\frac{v_1}{\delta} (z\delta) \right)^3 + \dots \right] - \frac{v_1}{u_2} \\
&= \frac{v_1}{u_2} \left[1 + \frac{\delta^2}{v_1} z + \frac{\delta^4}{3v_1} z^3 + \dots \right] \left[1 + v_1 z + v_1^2 z^2 + \left(\frac{v_1 \delta^2}{3} + v_1^3 \right) z^3 + \dots \right] - \frac{v_1}{u_2} \\
&= u_0 z + u_0(u_1 + u'_1) \frac{z^2}{2!} + u_0((u_1 + u'_1)^2 + 2u_0 u_2) \frac{z^3}{3!} + \dots
\end{aligned}$$

The same coefficients could also be recovered directly from the differential equation by repeated differentiation, but the closed form is shorter.

Step 3. Check the first few objects directly.

There is also a direct check: list the objects for small n .

$$\begin{aligned}
\mathcal{P}_1 &= \{\textcircled{1}\} \implies \\
f_1(\mathbf{u}) &= u_0 \\
\mathcal{P}_2 &= \{\textcircled{1}-\textcircled{2}, \textcircled{2}-\textcircled{1}\} \implies \\
f_2(\mathbf{u}) &= u_1 u_0 + u_0 u'_1 = u_0(u_1 + u'_1) \\
\mathcal{P}_3 &= \{\textcircled{1}-\textcircled{2}-\textcircled{3}, \textcircled{1}-\textcircled{3}-\textcircled{2}, \textcircled{2}-\textcircled{1}-\textcircled{3}, \textcircled{2}-\textcircled{3}-\textcircled{1}, \textcircled{3}-\textcircled{1}-\textcircled{2}, \textcircled{3}-\textcircled{2}-\textcircled{1}\} \implies \\
f_3(\mathbf{u}) &= u_1 u_1 u_0 + u_1 u_0 u'_1 + u_0 u_2 u_0 + u_1 u_0 u'_1 + u_0 u_2 u_0 + u_0 u'_1 u'_1 \\
&= u_0(u_1^2 + u_1 u'_1 + u_0 u_2 + u_1 u'_1 + u_0 u_2 + u_1'^2) \\
&= u_0((u_1 + u'_1)^2 + 2u_0 u_2) \\
\mathcal{P}_4 &= \{\dots\} \\
f_4(\mathbf{u}) &= u_0(u_1 + u'_1)((u_1 + u'_1)^2 + 8u_0 u_2)
\end{aligned}$$

Indeed, if $a = u_1 + u'_1$ and $\hat{I}(z, \mathbf{u}) = \sum f_n(\mathbf{u}) z^n / n!$, the differential equation gives

$$f_4 = a f_3 + u_2 (3f_1 f_2 + 3f_2 f_1) = a u_0 (a^2 + 2u_0 u_2) + 6a u_0^2 u_2 = a u_0 (a^2 + 8u_0 u_2).$$

With the closed form established, we can now extract the mean numbers of the three node types by specializing one marker at a time.

Nullary Type

Set $u_1 = u'_1 = u_2 = 1$. Then $v_1 = 1$ and $\delta = \sqrt{u_0 - 1}$. We have

$$\hat{I}(z, u_0) = \delta \frac{1 + \delta \tan(z\delta)}{\delta - \tan(z\delta)} - 1 = (1 + \delta^2) \frac{w}{1 - w}$$

where $w := \frac{\tan(z\delta)}{\delta} = z + \frac{z^3}{3}\delta^2 + O(\delta^4)$. Write $w = w_0 + w_2\delta^2 + O(\delta^4)$ with $w_0 = z$ and $w_2 = z^3/3$. Then

$$1 - w = 1 - w_0 - w_2\delta^2 + O(\delta^4) = (1 - w_0) \left[1 - \frac{w_2}{1 - w_0}\delta^2 + O(\delta^4) \right]$$

Hence,

$$\frac{1}{1 - w} = \frac{1}{1 - w_0} \frac{1}{1 - \frac{w_2}{1 - w_0}\delta^2 + \dots} = \frac{1}{1 - w_0} \left[1 + \frac{w_2}{1 - w_0}\delta^2 + O(\delta^4) \right]$$

Multiplying by $w = w_0 + w_2\delta^2 + O(\delta^4)$ yields:

$$\frac{w}{1 - w} = \frac{w_0}{1 - w_0} + \frac{w_2}{(1 - w_0)^2}\delta^2 + O(\delta^4)$$

Substituting $w_0 = z$ and $w_2 = z^3/3$, we obtain

$$\frac{w}{1 - w} = \frac{z}{1 - z} + \frac{z^3}{3(1 - z)^2}\delta^2 + O(\delta^4) \quad (3.16)$$

Since $\delta^2 = u_0 - 1$, the first-order Taylor term in u_0 is exactly the coefficient of δ^2 . Hence,

$$\left. \frac{\partial I}{\partial u_0} \right|_{u_0=1} = [\delta^2] \left((1 + \delta^2) \frac{w}{1 - w} \right) = \frac{z}{1 - z} + \frac{z^3}{3(1 - z)^2}$$

For $n \geq 3$, the coefficient of z^n in $\left. \frac{\partial I}{\partial u_0} \right|_{u_0=1}$ is $(n + 1)/3$.

Unary Type

Treat the two unary types together. Note that $\hat{I}$ depends on u_1, u'_1 only through $v_1 = (u_1 + u'_1)/2$. Set $u_0 = u_2 = 1$. Then $\delta = \sqrt{1 - v_1^2}$ and

$$\hat{I}(z, v_1) = \delta \frac{v_1 + \delta \tan(z\delta)}{\delta - v_1 \tan(z\delta)} - v_1 = \frac{v_1 + \delta^2 w}{1 - v_1 w} - v_1 = \frac{w}{1 - v_1 w}$$

where $w := \frac{\tan(z\delta)}{\delta} = z + \frac{z^3}{3}\delta^2 + O(\delta^4)$. We have

$$\frac{\partial I}{\partial u_1} = \frac{\partial I}{\partial u'_1} = \frac{1}{2} \frac{\partial I}{\partial v_1}$$

So it is enough to compute $\frac{\partial I}{\partial v_1}$ at $v_1 = 1$. First, w depends on v_1 through $s := \delta^2 = 1 - v_1^2$:

$$\frac{\partial w}{\partial v_1} = \frac{\partial w}{\partial s} \cdot (-2v_1)$$

From the series $\left. \frac{\partial w}{\partial s} \right|_{s=0} = \frac{z^3}{3}$, so at $v_1 = 1$:

$$\left. \frac{\partial w}{\partial v_1} \right|_{v_1=1} = -\frac{2z^3}{3}$$

Now differentiate $\hat{I}(z, v_1)$:

$$\frac{\partial \hat{I}}{\partial v_1} = \frac{\frac{\partial w}{\partial v_1}(1 - v_1 w) + w(w + v_1 \frac{\partial w}{\partial v_1})}{(1 - v_1 w)^2}$$

Substituting $v_1 = 1, w = z$ and $\frac{\partial w}{\partial v_1} = -\frac{2z^3}{3}$, we obtain

$$\left. \frac{\partial \hat{I}}{\partial v_1} \right|_{v_1=1} = \frac{z^2 (1 - \frac{2}{3}z)}{(1-z)^2}$$

Consequently

$$\left. \frac{\partial \hat{I}}{\partial u_1} \right|_{u_1=1} = \left. \frac{\partial \hat{I}}{\partial u'_1} \right|_{u'_1=1} = \frac{z^2 (1 - \frac{2}{3}z)}{2(1-z)^2}$$

For $n \geq 3$, the coefficient of z^n in $\left. \frac{\partial \hat{I}}{\partial u_1} \right|_{u_1=1}$ and $\left. \frac{\partial \hat{I}}{\partial u'_1} \right|_{u'_1=1}$ is $(n+1)/6$.

Binary Type

Set $u_0 = 1, v_1 = (1+1)/2 = 1$. Then $\delta = \sqrt{u_2 - 1}$ and

$$\hat{I}(z, u_2) = \frac{\delta}{u_2} \frac{1 + \delta \tan(z\delta)}{\delta - \tan(z\delta)} - \frac{1}{u_2} = \frac{w}{1-w}$$

where $w := \frac{\tan(z\delta)}{\delta} = z + \frac{z^3}{3}\delta^2 + O(\delta^4)$. This is the same calculation as in the nullary case, with one parameter renamed. From (3.16) we obtain

$$\left. \frac{\partial \hat{I}}{\partial u_2} \right|_{u_2=1} = [\delta^2] \frac{w}{1-w} = \frac{z^3}{3(1-z)^2}$$

For $n \geq 3$, the coefficient of z^n in $\left. \frac{\partial \hat{I}}{\partial u_2} \right|_{u_2=1}$ is $\frac{n-2}{3}$.

Summing the contributions of all node types:

$$\frac{n+1}{3} + \frac{n+1}{6} + \frac{n+1}{6} + \frac{n-2}{3} = n$$

as expected.

III.7.4. Symbolic inclusion-exclusion.

To make the relation $\mathcal{Q} \cong \mathcal{U} \star \mathcal{P}$ concrete, let us list the elements of $\mathcal{Q}_3$. Here $\mathcal{U}$ records the distinguished fixed points, while $\mathcal{P}$ records the permutation induced on the remaining labels. Thus an object of $\mathcal{U}_k \star \mathcal{P}_{3-k}$ is obtained by choosing k labels to be underlined fixed points, and then permuting the remaining $3-k$ labels.

$$\mathcal{U} = \{\emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\}, \dots\}$$

$$\mathcal{P} = \{\emptyset, (1), (1, 2), (2, 1), (1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1), \dots\}$$

Pick an element of $\mathcal{U}$ of size k and an element of $\mathcal{P}$ of size $3-k$, with $0 \leq k \leq 3$.

$$\mathcal{U}_0 \star \mathcal{P}_3$$

- $\emptyset \star (1, 2, 3)$: $\binom{3}{0} = 1$ way of relabelling
- $(\emptyset, (1, 2, 3)) \leftrightarrow (1, 2, 3)$

- $\emptyset \star (1, 3, 2)$: $\binom{3}{0} = 1$ way of relabelling
 - $(\emptyset, (1, 3, 2)) \leftrightarrow (1, 3, 2)$
- $\emptyset \star (2, 1, 3)$: $\binom{3}{0} = 1$ way of relabelling
 - $(\emptyset, (2, 1, 3)) \leftrightarrow (2, 1, 3)$
- $\emptyset \star (2, 3, 1)$: $\binom{3}{0} = 1$ way of relabelling
 - $(\emptyset, (2, 3, 1)) \leftrightarrow (2, 3, 1)$
- $\emptyset \star (3, 1, 2)$: $\binom{3}{0} = 1$ way of relabelling
 - $(\emptyset, (3, 1, 2)) \leftrightarrow (3, 1, 2)$
- $\emptyset \star (3, 2, 1)$: $\binom{3}{0} = 1$ way of relabelling
 - $(\emptyset, (3, 2, 1)) \leftrightarrow (3, 2, 1)$

$\mathcal{U}_1 \star \mathcal{P}_2$

- $\{1\} \star (1, 2)$: $\binom{3}{1} = 3$ ways of relabelling
 - $(\{1\}, (2, 3)) \leftrightarrow (\underline{1}, 2, 3)$
 - $(\{2\}, (1, 3)) \leftrightarrow (1, \underline{2}, 3)$
 - $(\{3\}, (1, 2)) \leftrightarrow (1, 2, \underline{3})$
- $\{1\} \star (2, 1)$: $\binom{3}{1} = 3$ ways of relabelling
 - $(\{1\}, (3, 2)) \leftrightarrow (\underline{1}, 3, 2)$
 - $(\{2\}, (3, 1)) \leftrightarrow (3, \underline{2}, 1)$
 - $(\{3\}, (2, 1)) \leftrightarrow (2, 1, \underline{3})$

$\mathcal{U}_2 \star \mathcal{P}_1$

- $\{1, 2\} \star (1)$: $\binom{3}{2} = 3$ ways of relabelling
 - $(\{1, 2\}, (3)) \leftrightarrow (\underline{1}, \underline{2}, 3)$
 - $(\{1, 3\}, (2)) \leftrightarrow (\underline{1}, 2, \underline{3})$
 - $(\{2, 3\}, (1)) \leftrightarrow (1, \underline{2}, \underline{3})$

$\mathcal{U}_3 \star \mathcal{P}_0$

- $\{1, 2, 3\} \star \emptyset$: $\binom{3}{3} = 1$ way of relabelling
 - $(\{1, 2, 3\}, \emptyset) \leftrightarrow (\underline{1}, \underline{2}, \underline{3})$

The count also matches:

$$|\mathcal{Q}_3| = \sum_{k=0}^3 \binom{3}{k} (3-k)! = 6 + 6 + 3 + 1 = 16.$$

Example III.25. *Rises and ascending runs in permutations.*

Rises

For rises, inclusion–exclusion is applied to permutations with distinguished rises. Consecutive distinguished rises form a cluster. A cluster with m letters contains $m - 1$ distinguished rises, so its EGF is

$$C(z, v) = \sum_{m \geq 2} v^{m-1} \frac{z^m}{m!} = \frac{e^{zv} - 1 - zv}{v}.$$

A permutation with distinguished rises is then a sequence of isolated atoms and clusters:

$$B(z, v) = \frac{1}{1 - z - C(z, v)} = \frac{v}{v + 1 - e^{zv}}.$$

If $A(z, u)$ marks all rises, then $B(z, v) = A(z, 1 + v)$, hence

$$A(z, u) = B(z, u - 1) = \frac{u - 1}{u - e^{z(u-1)}}.$$

Step 1. Expand around $z = 0$.

First expand $A(z, u)$ as a power series in z around 0.

$$A(z, u) = \frac{u - 1}{u - e^{z(u-1)}}$$

The denominator is:

$$\begin{aligned} u - e^{z(u-1)} &= u - \left[1 + (u-1)z + \frac{(u-1)^2}{2}z^2 + \frac{(u-1)^3}{6}z^3 + \frac{(u-1)^4}{24}z^4 + O(z^5) \right] \\ &= (u-1) - (u-1)z - \frac{(u-1)^2}{2}z^2 - \frac{(u-1)^3}{6}z^3 - \frac{(u-1)^4}{24}z^4 + O(z^5) \\ &= (u-1) \left[1 - z - \frac{u-1}{2}z^2 - \frac{(u-1)^2}{6}z^3 - \frac{(u-1)^3}{24}z^4 + O(z^5) \right] \end{aligned}$$

Hence,

$$\begin{aligned} A(z, u) &= \left(1 - z - \frac{u-1}{2}z^2 - \frac{(u-1)^2}{6}z^3 - \frac{(u-1)^3}{24}z^4 + O(z^5) \right)^{-1} \\ &= \left(1 - z - \frac{u-1}{2}z^2 - \frac{(u-1)^2}{6}z^3 - \frac{(u-1)^3}{24}z^4 \right)^{-1} + O(z^5) \end{aligned}$$

Set

$$w = z + \frac{u-1}{2}z^2 + \frac{(u-1)^2}{6}z^3 + \frac{(u-1)^3}{24}z^4$$

Compute powers of w up to order z^4 :

$$\begin{aligned}
 w^2 &= \left(z + \frac{u-1}{2}z^2 + \frac{(u-1)^2}{6}z^3 \right)^2 \\
 &= z^2 + \left(\frac{u-1}{2}z^2 \right)^2 + 2z \frac{u-1}{2}z^2 + 2z \frac{(u-1)^2}{6}z^3 \\
 &= z^2 + (u-1)z^3 + \frac{7}{12}(u-1)^2z^4 \\
 w^3 &= \left(z + \frac{u-1}{2}z^2 \right)^3 = z^3 + 3z^2 \frac{u-1}{2}z^2 = z^3 + \frac{3}{2}(u-1)z^4 \\
 w^4 &= z^4
 \end{aligned}$$

Then

$$\begin{aligned}
 A(z, u) &= 1 + z + \left(\frac{u-1}{2} + 1 \right) z^2 + \left(\frac{(u-1)^2}{6} + (u-1) + 1 \right) z^3 \\
 &\quad + \left(\frac{(u-1)^3}{24} + \frac{7}{12}(u-1)^2 + \frac{3}{2}(u-1) + 1 \right) z^4 + O(z^5) \\
 &= 1 + z + (u+1) \frac{z^2}{2} + (u^2 + 4u + 1) \frac{z^3}{6} + (u^3 + 11u^2 + 11u + 1) \frac{z^4}{24} + \dots
 \end{aligned}$$

Step 2. Expand around $u = 1$.

Now expand $A(z, u)$ in powers of u around $u = 1$. Set $v = u - 1$.

$$A(z, u) = \frac{v}{(v+1) - e^{zv}}$$

The denominator

$$v + 1 - e^{zv} = v + 1 - \left(1 + zv + \frac{z^2}{2}v^2 + \frac{z^3}{6}v^3 + O(v^4) \right) = (1-z)v - \frac{z^2}{2}v^2 - \frac{z^3}{6}v^3 + O(v^4)$$

Hence,

$$\begin{aligned}
 A(z, u) &= \left(1 - z - \frac{z^2}{2}v - \frac{z^3}{6}v^2 + O(v^3) \right)^{-1} \\
 &= \frac{1}{1-z} \left(1 - \frac{z^2}{2(1-z)}v - \frac{z^3}{6(1-z)}v^2 + O(v^3) \right)^{-1} \\
 &= \frac{1}{1-z} \left(1 - \frac{z^2}{2(1-z)}v - \frac{z^3}{6(1-z)}v^2 \right)^{-1} + O(v^3) \\
 &= \frac{1}{1-z} \left(1 + \frac{z^2}{2(1-z)}v + \frac{z^3}{6(1-z)}v^2 + \left(\frac{z^2}{2(1-z)}v \right)^2 \right) + O(v^3) \\
 &= \frac{1}{1-z} + \frac{z^2}{2(1-z)^2}v + \frac{2z^3 + z^4}{6(1-z)^3} \frac{v^2}{2!} + \dots
 \end{aligned}$$

Step 3. Extract the mean and variance.

In the expansion around $u = 1$, the coefficient of $(u - 1)$ is the cumulative GF for the first factorial moment, while the coefficient of $(u - 1)^2/2!$ is the cumulative GF for the second factorial moment. Extracting coefficients gives:

$$[z^n] \frac{z^2}{2(1-z)^2} = \frac{1}{2} [z^{n-2}] \sum_{m=0}^{\infty} (m+1) z^m = \frac{n-1}{2}$$

$$[z^n] \frac{2z^3 + z^4}{6(1-z)^3} = \frac{1}{3} [z^{n-3}] \sum_{m=0}^{\infty} \binom{m+2}{2} z^m + \frac{1}{6} [z^{n-4}] \sum_{m=0}^{\infty} \binom{m+2}{2} z^m = \frac{3n^2 - 11n + 10}{12}$$

The mean of the number of rises in a random permutation of size n is $\frac{n-1}{2}$, and for $n \geq 2$ the variance is:

$$\frac{3n^2 - 11n + 10}{12} + \frac{n-1}{2} - \left(\frac{n-1}{2} \right)^2 = \frac{n+1}{12}$$

Ascending Runs

Step 1. Fix the indexing first.

Here it is safer not to use the word “length” too casually. I will say that an ascending run has size r if it consists of r consecutive entries

$$\sigma_i < \sigma_{i+1} < \cdots < \sigma_{i+r-1}.$$

Thus a rise has size 2, but it has one adjacent comparison. With this convention, a cluster of overlapping distinguished ascending runs has OGF

$$I(z, v) = \frac{vz^r}{1 - v(z + z^2 + \cdots + z^{r-1})}.$$

The numerator z^r gives the first distinguished run. Each new overlapping run may start $1, 2, \dots, r-1$ positions to the right of the previous one, which explains the denominator. After the ordinary-to-exponential transform, permutations with distinguished runs are counted by

$$B(z, v) = \frac{1}{1 - z - \widehat{I}(z, v)}, \quad A(z, u) = B(z, u - 1).$$

Step 2. Check the partial fraction statement.

The rational function $I(z, v)$ is improper: the numerator has degree r , while the denominator has degree $r-1$. For $r \geq 3$, polynomial division gives

$$I(z, v) = 1 - z + \frac{(v+1)z - 1}{1 - v(z + z^2 + \cdots + z^{r-1})}.$$

Therefore the transformed form should be

$$\widehat{I}(z, v) = 1 - z + \sum_{j=1}^{r-1} c_j(v) e^{\omega_j(v)z},$$

where the $\omega_j(v)$ are the roots of

$$\omega^{r-1} = v(1 + \omega + \cdots + \omega^{r-2}).$$

The factor $(1 - z)(v + 1)$ printed in the book does not match this division. The special case $r = 2$ is also a good sanity check: then

$$I(z, v) = \frac{vz^2}{1 - vz} = -z - \frac{1}{v} + \frac{1/v}{1 - vz},$$

so

$$\widehat{I}(z, v) = \frac{e^{vz} - 1 - vz}{v},$$

which is exactly the cluster series used above for rises.

Step 3. Estimate the moments.

Let $X_{n,r}$ be the number of ascending runs of size r in a random permutation of size n . There are $n - r + 1$ possible windows, and a fixed window of r entries is increasing with probability $1/r!$. Hence, exactly,

$$\mathbb{E}X_{n,r} = \frac{n - r + 1}{r!}.$$

For the variance, write $X_{n,r} = \sum_i Y_i$, where Y_i marks the window starting at i . Disjoint windows are independent as far as their internal relative orders are concerned. If two windows start d positions apart, with $1 \leq d \leq r - 1$, then both are increasing iff the whole block of $r + d$ entries is increasing. Thus

$$\mathbb{P}(Y_i = Y_{i+d} = 1) = \frac{1}{(r + d)!}.$$

Consequently,

$$\begin{aligned} \text{Var}(X_{n,r}) &= (n - r + 1) \frac{1}{r!} \left(1 - \frac{1}{r!} \right) \\ &\quad + 2 \sum_{d=1}^{r-1} (n - r + 1 - d) \left(\frac{1}{(r + d)!} - \frac{1}{(r!)^2} \right). \end{aligned}$$

In particular $\text{Var}(X_{n,r}) = O(n/r!)$ in the threshold range we care about. Therefore

$$\frac{\text{Var}(X_{n,r})}{(\mathbb{E}X_{n,r})^2} = O\left(\frac{r!}{n}\right),$$

so Chebyshev gives concentration as soon as $r! = o(n)$. This is the natural condition for runs of size r . In the book's later wording, the mean is written as

$$\frac{n - \ell + 1}{\ell!},$$

which corresponds to our $r = \ell$. The displayed condition $(\ell + 1)! = o(n)$ is stronger than what this variance estimate needs; it is not the sharp threshold condition suggested by the preceding mean estimate.

▷ III.36. Permutations without ℓ -ascending runs.

Here the book uses ℓ in the comparison-length sense: an ℓ -ascending run has $\ell + 1$ entries. To avoid mixing this with the size convention above, set $m = \ell + 1$. Then excluding ℓ -ascending runs is the same as excluding increasing windows of size m .

Consider the denominator of $A(z, u)|_{u=0}$ before applying the ordinary-to-exponential transform. From the cluster OGF with $r = m$ and $v = -1$,

$$1 - z - I(z, -1) = 1 - z + \frac{z^m}{1 + z + \dots + z^{m-1}} = \frac{1}{1 + z + \dots + z^{m-1}} = \frac{1 - z}{1 - z^m}.$$

To prepare for the transform $(1 - \omega z)^{-1} \mapsto e^{\omega z}$, decompose $\frac{1-z}{1-z^m}$ using the m -th roots of unity. Let $\omega_k = e^{2\pi i k/m}$ for $k = 0, 1, \dots, m-1$. The standard partial fraction expansion is:

$$\frac{1}{1 - z^m} = \frac{1}{m} \sum_{k=0}^{m-1} \frac{1}{1 - \omega_k z}.$$

Now apply the ordinary-to-exponential transform:

$$\begin{aligned} \text{EGF Denominator} &= \widehat{\frac{1 - z}{1 - z^m}} \\ &= \widehat{\frac{1}{1 - z^m}} - \widehat{\frac{z}{1 - z^m}} \\ &= \widehat{\frac{1}{1 - z^m}} - \int_0^z \widehat{\frac{1}{1 - t^m}} dt \\ &= \frac{1}{m} \sum_{k=0}^{m-1} e^{\omega_k z} - \int_0^z \frac{1}{m} \sum_{k=0}^{m-1} e^{\omega_k t} dt \\ &= \sum_{i \geq 0} \frac{z^{mi}}{(mi)!} - \sum_{i \geq 0} \frac{z^{mi+1}}{(mi+1)!}. \end{aligned}$$

At this point the root-of-unity filter has done its job: $\sum_{k=0}^{m-1} \omega_k^n$ vanishes unless $m \mid n$, in which case it equals m . The other small trick is that multiplying by z in the ordinary series becomes integration after passing to exponential generating functions.

Therefore the EGF of permutations without ℓ -ascending runs is

$$\left(\sum_{i \geq 0} \frac{z^{(\ell+1)i}}{((\ell+1)i)!} - \sum_{i \geq 0} \frac{z^{(\ell+1)i+1}}{((\ell+1)i+1)!} \right)^{-1}.$$

For orientation, here are the first few EGFs when $m = 2, 3, 4$:

$$\begin{aligned} \text{A.R. of length 1} \quad & \left(\sum_{i \geq 0} \frac{z^{2i}}{(2i)!} - \frac{z^{2i+1}}{(2i+1)!} \right)^{-1} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + O(z^5) \\ \text{A.R. of length 2} \quad & \left(\sum_{i \geq 0} \frac{z^{3i}}{(3i)!} - \frac{z^{3i+1}}{(3i+1)!} \right)^{-1} = 1 + z + 2\frac{z^2}{2!} + 5\frac{z^3}{3!} + 17\frac{z^4}{4!} + O(z^5) \\ \text{A.R. of length 3} \quad & \left(\sum_{i \geq 0} \frac{z^{4i}}{(4i)!} - \frac{z^{4i+1}}{(4i+1)!} \right)^{-1} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + 23\frac{z^4}{4!} + O(z^5) \end{aligned}$$

For example, there are 17 permutations in $\mathcal{P}_4$ without ascending runs of length 2: 1324, 1423, 1432, 2143, 2314, 2413, 2431, 3142, 3214, 3241, 3412, 3421, 4132, 4213, 4231, 4312, 4321.

▷ **III.41.** *Patterns in trees II.*

Let $\mathbf{t}$ be the fixed non-empty pattern, with $m = |\mathbf{t}|$. For a non-empty binary tree $\tau = (\tau_0, \tau_1)$, each occurrence of $\mathbf{t}$ is accounted for in one of three places: the left subtree, the right subtree, or the root itself when $\tau = \mathbf{t}$. Thus

$$\omega[\tau] = \omega[\tau_0] + \omega[\tau_1] + \llbracket \tau = \mathbf{t} \rrbracket, \quad \omega[\emptyset] = 0.$$

The only small trick is the following identity:

$$u^{\llbracket \tau = \mathbf{t} \rrbracket} u^{\omega[\tau_0]} u^{\omega[\tau_1]} = u^{\omega[\tau_0]} u^{\omega[\tau_1]} + \llbracket \tau = \mathbf{t} \rrbracket \cdot (u - 1)$$

Indeed, there are two cases.

- $\tau \neq \mathbf{t}$: LHS = $u^0 u^{\omega[\tau_0]} u^{\omega[\tau_1]} = u^{\omega[\tau_0]} u^{\omega[\tau_1]}$; RHS = $u^{\omega[\tau_0]} u^{\omega[\tau_1]} + 0 \cdot (u - 1) = u^{\omega[\tau_0]} u^{\omega[\tau_1]}$
- $\tau = \mathbf{t}$: the two proper subtrees of $\mathbf{t}$ are smaller than $\mathbf{t}$, so $\omega[\tau_0] = \omega[\tau_1] = 0$. Hence LHS = u and RHS = $1 + (u - 1) = u$.

Now sum over all binary trees τ to get the bivariate generating function:

$$\begin{aligned} B(z, u) &= \sum_{\tau} z^{|\tau|} u^{\omega[\tau]} \\ &= z^{|\emptyset|} u^{\omega[\emptyset]} + \sum_{\tau \neq \emptyset} z^{|\tau|} u^{\omega[\tau]} \\ &= z^0 u^0 + \sum_{\tau \neq \emptyset} z^{|\tau|} u^{\omega[\tau]} \\ &= 1 + \sum_{\tau \neq \emptyset} z^{|\tau|} (u^{\omega[\tau_0]} u^{\omega[\tau_1]} + \llbracket \tau = \mathbf{t} \rrbracket \cdot (u - 1)) \\ &= 1 + \sum_{\tau_0, \tau_1} z^{1+|\tau_0|+|\tau_1|} u^{\omega[\tau_0]} u^{\omega[\tau_1]} + \sum_{\tau \neq \emptyset} z^{|\tau|} \llbracket \tau = \mathbf{t} \rrbracket \cdot (u - 1) \\ &= 1 + z \sum_{\tau_0, \tau_1} z^{|\tau_0|} u^{\omega[\tau_0]} \cdot z^{|\tau_1|} u^{\omega[\tau_1]} + z^{|\mathbf{t}|} \cdot (u - 1) \\ &= 1 + zB(z, u)^2 + z^m(u - 1) \end{aligned}$$

Solving the quadratic and choosing the branch with $B(0, u) = 1$ gives

$$B(z, u) = \frac{1 - \sqrt{1 - 4z - 4(u - 1)z^{m+1}}}{2z}.$$

3.8 Extremal parameters

III.8.3. Averages and moments.

Let $f^{[h]}(z)$ be the GF of objects whose extremal parameter is at most h , and set $f^{[-1]}(z) = 0$. Then $f^{[h]}(z) - f^{[h-1]}(z)$ counts objects whose parameter is exactly h . To see why the first sum rearranges into the second, use a discrete summation-by-parts trick. The rearrangement is coefficientwise: for a fixed object size, only finitely many values of

h can contribute, so no analytic convergence issue is being hidden here. Write the linear factor h as

$$h = \sum_{j=1}^h 1$$

Substituting this produces a double sum:

$$\Xi(z) = \sum_{h=0}^{\infty} \left(\sum_{j=1}^h 1 \right) [f^{[h]}(z) - f^{[h-1]}(z)] = \sum_{h=1}^{\infty} \sum_{j=1}^h [f^{[h]}(z) - f^{[h-1]}(z)]$$

The summation region is $1 \leq j \leq h < \infty$. Reverse the order: then j runs from 1 to ∞ , while h runs from j to ∞ .

$$\Xi(z) = \sum_{j=1}^{\infty} \sum_{h=j}^{\infty} [f^{[h]}(z) - f^{[h-1]}(z)]$$

The inner sum over h now telescopes:

$$\sum_{h=j}^{\infty} [f^{[h]}(z) - f^{[h-1]}(z)] = \lim_{N \rightarrow \infty} \sum_{h=j}^N [f^{[h]}(z) - f^{[h-1]}(z)]$$

Expanding the finite sum makes the cancellation explicit:

$$\lim_{N \rightarrow \infty} ([f^{[j]} - f^{[j-1]}] + [f^{[j+1]} - f^{[j]}] + \cdots + [f^{[N]} - f^{[N-1]}]) = \lim_{N \rightarrow \infty} (f^{[N]}(z) - f^{[j-1]}(z))$$

Because $f^{[N]}(z)$ counts objects with parameter at most N , letting $N \rightarrow \infty$ recovers the unconstrained generating function $f(z)$. So the inner sum collapses to

$$f(z) - f^{[j-1]}(z)$$

Putting this back into the outer sum,

$$\Xi(z) = \sum_{j=1}^{\infty} [f(z) - f^{[j-1]}(z)]$$

Finally shift the index by $h = j - 1$:

$$\Xi(z) = \sum_{h=0}^{\infty} [f(z) - f^{[h]}(z)]$$

3.9 Perspective

Chapter 4

COMPLEX ANALYSIS, RATIONAL AND MEROMORPHIC ASYMPTOTICS

4.1 Generating functions as analytic objects

▷ **IV.6.** *Integrals of rational fractions.*

We compute the two displayed integrals by closing the contour in the upper half-plane. The only work is then residue computation plus a little finite algebra.

$$J_m = \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^m}$$

Set

$$f(z) = \frac{1}{(1+z^2)^m} = \frac{1}{(z-i)^m(z+i)^m}.$$

Integrate over the contour Γ_R formed by the real interval $[-R, R]$ and the upper semicircle $C_R = \{z = Re^{i\theta} : 0 \leq \theta \leq \pi\}$.

For $z \in C_R$ and $R > 1$, $|z^2 + 1| \geq |z^2| - 1 = R^2 - 1$. Thus

$$|f(z)| \leq \frac{1}{(R^2 - 1)^m}$$

The length of C_R is πR , so

$$\left| \int_{C_R} f(z) dz \right| \leq \frac{\pi R}{(R^2 - 1)^m} \rightarrow 0 \text{ as } R \rightarrow \infty$$

So the contour integral reduces to a residue computation at the pole in the upper half-plane. Therefore

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} f(z) dz = \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^m} = J_m$$

For $R > 1$, the only singularity inside Γ_R is the pole of order m at $z = i$. The residue theorem then gives:

$$\int_{\Gamma_R} f(z) dz = 2\pi i \operatorname{Res}(f; i)$$

Now compute the residue at $z = i$:

$$\operatorname{Res}(f; i) = \frac{1}{(m-1)!} \lim_{z \rightarrow i} \frac{d^{m-1}}{dz^{m-1}} [(z-i)^m f(z)] = \frac{1}{(m-1)!} \lim_{z \rightarrow i} \frac{d^{m-1}}{dz^{m-1}} (z+i)^{-m}$$

The k th derivative of $g(z) := (z+i)^{-m}$ is:

$$g^{(k)}(z) = (-1)^k \frac{(m+k-1)!}{(m-1)!} (z+i)^{-m-k}$$

For $k = m-1$:

$$g^{(m-1)}(z) = (-1)^{m-1} \frac{(2m-2)!}{(m-1)!} (z+i)^{-2m+1}$$

Evaluating at $z = i$ gives

$$g^{(m-1)}(i) = (-1)^{m-1} \frac{(2m-2)!}{(m-1)!} (2i)^{-2m+1}$$

Therefore,

$$\begin{aligned} \operatorname{Res}(f; i) &= \frac{1}{(m-1)!} (-1)^{m-1} \frac{(2m-2)!}{(m-1)!} (2i)^{-2m+1} \\ &= (-1)^{m-1} \frac{(2m-2)!}{[(m-1)!]^2} \frac{(-1)^m i}{2^{2m-1}} \\ &= -i \frac{(2m-2)!}{[(m-1)!]^2 2^{2m-1}} \\ J_m &= 2\pi i \left(-i \frac{(2m-2)!}{[(m-1)!]^2 2^{2m-1}} \right) = \frac{\pi}{2^{2m-2}} \binom{2m-2}{m-1} = \frac{\pi}{4^{m-1}} \binom{2m-2}{m-1} \end{aligned} \quad (4.1)$$

$$K_m = \int_{-\infty}^{\infty} \frac{dx}{(1^2 + x^2)(2^2 + x^2) \cdots (m^2 + x^2)}$$

Now set

$$f(z) = \frac{1}{\prod_{k=1}^m (z^2 + k^2)}.$$

We use the same contour Γ_R .

For $R > m$, the poles inside Γ_R are $z_k = ik, k = 1, \dots, m$. On C_R , $|z^2 + k^2| \geq R^2 - k^2 \geq (R/2)^2$ eventually, so $|f(z)| \leq \text{const}/R^{2m}$ while the length of C_R is πR . Hence,

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$$

and the residue theorem yields

$$K_m = 2\pi i \sum_{k=1}^m \operatorname{Res}(f; ik)$$

Since the poles are simple,

$$\operatorname{Res}(f; ik) = \lim_{z \rightarrow ik} (z - ik) f(z) = \frac{1}{2ik} \prod_{j \neq k} \frac{1}{(ik)^2 + j^2} = \frac{1}{2ik} \prod_{j \neq k} \frac{1}{j^2 - k^2}$$

Therefore,

$$K_m = 2\pi i \sum_{k=1}^m \frac{1}{2ik} \prod_{j \neq k} \frac{1}{j^2 - k^2} = \pi \sum_{k=1}^m \frac{1}{k} \prod_{j \neq k} \frac{1}{j^2 - k^2}$$

The analytic work is now done; what remains is straightforward algebra, though it takes a little patience. Factor the product:

$$\prod_{j \neq k} (j^2 - k^2) = \prod_{j \neq k} (j - k) \prod_{j \neq k} (j + k).$$

For fixed k ,

$$\prod_{j \neq k} (j - k) = (1 - k)(2 - k) \cdots ((k - 1) - k)(1)(2) \cdots (m - k) = (-1)^{k-1} (k - 1)! (m - k)!$$

$$\prod_{j \neq k} (j + k) = (1 + k)(2 + k) \cdots ((k - 1) + k)(2k + 1)(2k + 2) \cdots (m + k) = \frac{(m + k)!}{k!(2k)!}$$

Therefore,

$$\frac{1}{k} \prod_{j \neq k} \frac{1}{j^2 - k^2} = \frac{1}{k} \left((-1)^{k-1} (k - 1)! (m - k)! \cdot \frac{(m + k)!}{k!(2k)!} \right)^{-1} = \frac{(-1)^{k-1} 2k}{(m - k)! (m + k)!}$$

and

$$K_m = \pi \sum_{k=1}^m \frac{(-1)^{k-1} 2k}{(m - k)! (m + k)!} = \frac{2\pi}{(2m)!} \sum_{k=1}^m (-1)^{k-1} k \binom{2m}{m - k}$$

To simplify the sum

$$S := \sum_{k=1}^m (-1)^{k-1} k \binom{2m}{m - k},$$

generating functions are again the quickest tool. The aim is to turn the alternating sum into a single coefficient extraction. Recall that the coefficient of z^m in a product $A(z)B(z)$ is

$$[z^m]A(z)B(z) = \sum_k a_k b_{m-k}$$

Take

$$A(z) = \sum_{k=1}^m (-1)^{k-1} k z^k, \quad B(z) = (1 + z)^{2m} = \sum_{j=0}^{2m} \binom{2m}{j} z^j.$$

Then the coefficient of z^m in $A(z)B(z)$ is exactly S :

$$S = [z^m] \sum_{k=1}^m (-1)^{k-1} k z^k (1 + z)^{2m} \tag{4.2}$$

Since we only care about the coefficient of z^m , terms of degree $> m$ in $A(z)$ are irrelevant. So we may extend the sum to infinity without changing the coefficient:

$$A(z) \equiv \sum_{k=1}^{\infty} (-1)^{k-1} k z^k = \frac{z}{(1 + z)^2} \pmod{z^{m+1}}.$$

Hence, by (4.2),

$$S = [z^m] \frac{z}{(1+z)^2} \cdot (1+z)^{2m} = [z^m] z(1+z)^{2m-2} = \binom{2m-2}{m-1}$$

Finally,

$$K_m = \frac{2\pi}{(2m)!} \binom{2m-2}{m-1} = \frac{\pi}{m(2m-1)[(m-1)!]^2} \quad (4.3)$$

4.2 Analytic functions and meromorphic functions

4.3 Singularities and exponential growth of coefficients

4.4 Closure properties and computable bounds

4.5 Rational and meromorphic functions

4.6 Localization of singularities

4.7 Singularities and functional equations

4.8 Perspective